



Smart Grids Group 2 Final Report

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Contents

1	Introduction	3
1.1	Background	3
1.2	Overview	3
2	Base Case	3
2.1	Network Configuration	3
2.2	Time-series Simulation Approach	5
2.3	Results Analysis Method	5
2.3.1	Safe Operation	5
2.3.2	Reliability Modeling	6
2.3.3	Cost Analysis	6
3	Base Case Results and Diagnosis	6
3.1	Bus Voltages	7
3.2	Demand Coverage	7
3.3	Equipment Loading	8
3.4	Reliability Impact	8
3.5	Annual Operating Cost	9
3.6	Diagnosis for Grid Upgrades	9
4	Phase One Upgrades	10
4.1	N-1 Criterion Upgrade	10
4.2	Voltage Performance	10
4.3	Demand Coverage & Imports	11
4.4	Equipment Loading	12
4.5	Line Losses	13
4.6	Reliability Costs	13
4.7	Annual Operating Cost	14
4.8	Modifying Voltage Level	14
4.9	Phase One Diagnosis	16
5	Phase Two Upgrades	16
5.1	Solution 1: RES Integration	16
5.2	Phase 2 modifications: N-1 reinforcement and RES additions	17
5.2.1	Additional lines to meet N-1 criterion	17
5.3	Derivation of Installed RES Capacities (Falset PV and Conesa Wind)	17
5.3.1	Falset Solar PV Capacity	18
5.3.2	Conesa Wind Capacity	18
5.3.3	RES integration and motivation (mitigating Bus 13 undervoltage)	18
5.4	Time-series simulation and recorded outputs	19
5.5	Results and diagnosis (Phase 2, Solution 1)	19
5.5.1	Voltage profile and deviation assessment	19

5.5.2	Demand coverage, imports/exports, and energy balance	19
5.5.3	Equipment loading (lines and transformers)	20
5.5.4	Transmission line loss requirement ($< 2\%$)	21
5.5.5	Annual operating cost and failure-related penalties	21
5.6	Investment cost of RES (solar and wind plants)	21
5.7	Summary of findings (Solution 1 diagnosis)	22
5.8	Phase 2: Solution 2: Storage and HVDC	24
5.8.1	Energy Storage at Valls with HVDC Connection to Montblanc .	24
5.9	Results and diagnosis (Phase 2, Solution 2)	25
5.9.1	Voltage profile and deviation assessment	25
5.9.2	Demand coverage, imports/exports, and energy balance	25
5.9.3	Equipment loading (lines and transformers)	26
5.9.4	Transmission line loss requirement ($< 2\%$)	27
5.9.5	Energy Balance and Annual Costs	28
5.9.6	CAPEX for Energy Storage and HVDC Connection	28
5.10	Investment cost of BESS and HVDC connection	28
5.11	Summary of Findings (Solution 2 Diagnosis)	29
5.12	Solution 3: Demand Response	30
5.12.1	Methodology: Load Shifting Strategy and Constraints	30
5.12.2	Simulation Results	31
5.12.3	Conclusion and Recommendation for Solution 3	33
5.13	Data Formats and Standards	46
5.14	Key Considerations	46
6	Conclusion	48

1 Introduction

1.1 Background

Electric power systems are undergoing a rapid transition driven by decarbonization targets, falling costs of renewable generation, and the electrification of end-uses. This shift is changing the fundamental operating conditions of the grid: instead of a few large central generators supplying passive consumers, modern networks increasingly host distributed energy resources (DER), storage, and active consumers who can react to system needs. This transition requires improved sensing, monitoring, control, and coordination so that the system can remain reliable while integrating variable generation and more dynamic demand.

1.2 Overview

The project is divided between a classical grid analysis and an application of the Smart Grids Architecture Model (SGAM) to a specific hypothetical use case. The classical grid scenario is a simplified arrangement of a transmission area in southern Catalonia. A single nuclear plant supplies four different load centers, supplemented by a connection to an external grid which supplies the rest of the power demand at great cost. The example system is modeled in PandaPower and analyzed for its shortcomings in supplying demand, equipment stresses, and power import costs, forming a base case from which two phases of upgrades are implemented. The first phase establishes N-1 compliance of the network, and the second explores different solutions to reduce operating costs of the network.

For the SGAM application, Group II focused on the Agricultural Energy Hub scenario concerning the region of farmland surrounding Igualada. The application of smart grids technology aims to modernize irrigation through renewable-powered pumping and smart metering. A robust plan for information sharing allows farmers to schedule water and electricity use using price signals and weather forecasts, thereby reducing operating costs while improving efficiency.

2 Base Case

This chapter describes the study area, the existing electrical network model, and the methodology used to evaluate the current system performance over a 24-hour operating cycle.

2.1 Network Configuration

The study focuses on the area around Tàrraga in Catalonia, Spain, shown in the regional map in Fig. 1. The map shows the consumption buses and the transmission lines connecting them to the nuclear generation and the external grid. Unconnected buses

are shown as potential upgrades, discussed further in Sections 4 and 5. The lines are represented as single or double circuits, reflecting different transmission capacities and redundancy levels. This geographical overview provides the physical context for the electrical model used in the base-case assessment.

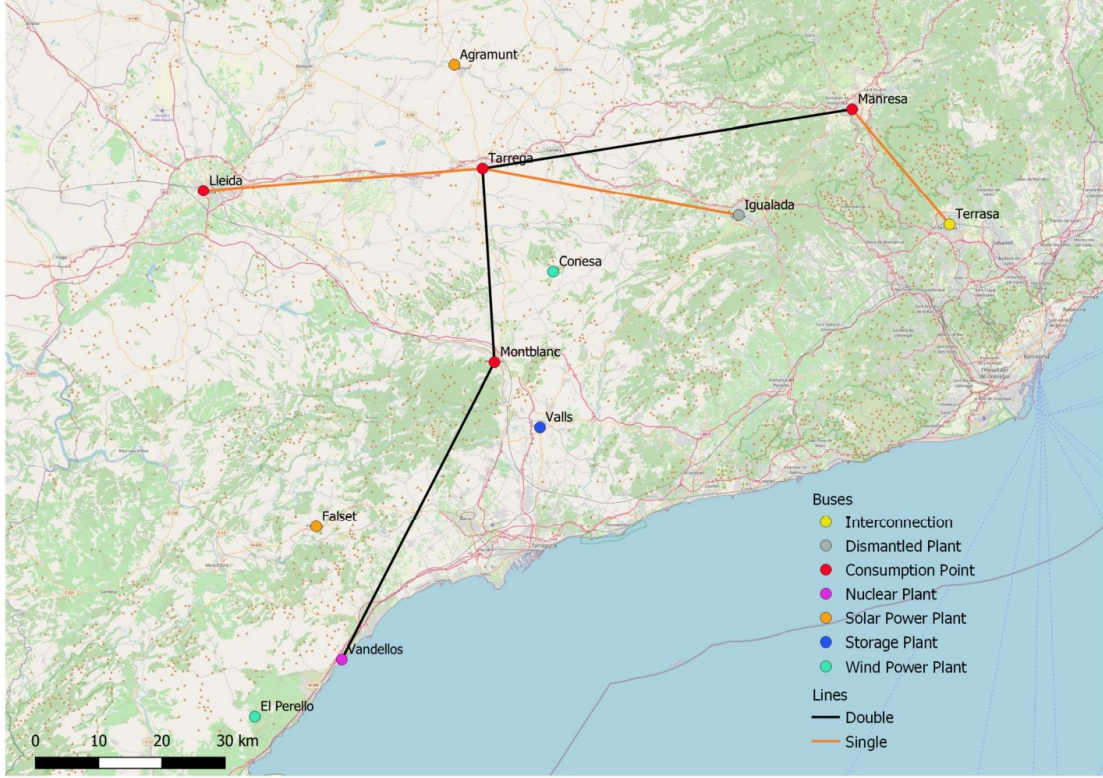


Figure 1: Region Map

The system is modeled in **PandaPower** based on given performance parameters for lines and transformers as well as rated values for loads and generators required for steady-state power-flow analysis. In the base case, the principal consumption points are represented as four loads located at:

1. Lleida (load 0, bus 15),
2. Montblanc (load 1, bus 14),
3. Manresa (load 2, bus 12), and
4. Tàrraga (load 3, bus 13).

Generation is represented by a nuclear unit modeled as a controllable generator (**gen**) in the base-case system, while the external grid (**ext_grid**) supplies the remaining demand and serves as the slack bus for the power flow simulation. The load buses are PQ with **0.98 power factor**, and the generator buses are PV with 1 p.u. and a specified power profile as detailed in the next section.

2.2 Time-series Simulation Approach

To reflect realistic daily operation, 24-hour profiles are used for demand and generation. The generation and consumption profiles are given by the Spanish national TSO [5], from which a representative working-day dataset gives time-series signals for demand and generation resources based on the national aggregation. This gives a representative curve for the total demand, solar generation, and wind generation over the course of a day. Each profile is first normalized by its maximum value to obtain per-unit curves, and then scaled by the nominal active power ratings defined in the network model. Concretely, for each load i , the hourly active power is computed as

$$P_{L,i}(t) = p_{L,i}^{\text{nom}} \cdot \alpha_L(t),$$

where $p_{L,i}^{\text{nom}}$ is the nominal load power at time t in the grid model and $\alpha_L(t) \in [0, 1]$ is the normalized demand profile. Similarly, the nuclear generation dispatch is defined as

$$P_G(t) = p_G^{\text{nom}} \cdot \alpha_G(t),$$

with p_G^{nom} taken from the generator rating and $\alpha_G(t)$ the normalized nuclear profile. This process is repeated in Section 5 for wind and solar generators using their respective profiles.

The time-series simulation is implemented using `pandapower.timeseries`. A profile table is created containing the hourly values for each load and the nuclear generator. These profiles are linked to network elements using `ConstControl` objects so that, at each hourly time step, the corresponding element setpoints (e.g., `p_mw`) are updated automatically. The simulation is executed by performing a normal power flow simulation at each hour's load and generation data point for a total of 24 results that are logged with `OutputWriter`.

2.3 Results Analysis Method

The base-case network is evaluated using the following performance indicators over the 24-hour period and, where required, extrapolated to annual figures.

2.3.1 Safe Operation

Bus voltage magnitudes are checked against acceptable operating bounds, where

$$V_{\min} = 0.90 \text{ p.u.}, \quad V_{\max} = 1.10 \text{ p.u.}$$

with violating buses being noted to inform upgrade decisions in the later phases of the project. Line and transformer loading are also reported in the simulation results. Any line or transformer exceeding the normal-operation threshold of 80% at any hour of the simulation are also identified as reinforcement needs in the current system.

2.3.2 Reliability Modeling

To quantify service continuity, an annual interruptibility assessment is performed using provided failure-rate models. For lines, the number of failures per year is estimated as a function of line length and maximum loading:

$$\lambda_{\text{line}} = 0.05 \cdot L_{\text{km}} + 0.005 \cdot \max(\text{Loading}_{\%}),$$

and for transformers:

$$\lambda_{\text{trafo}} = 0.15 + 0.002 \cdot \max(\text{Loading}_{\%}).$$

The annual failure rate is combined with assumed repair times of 2 hours and 8 hours per failure for lines and transformers, respectively to compute the annual downtime (hours/year) for each equipment item. Finally, to derive a figure for energy lost to failures used in the next sections' cost analysis, the nominal power is multiplied by the downtime. For the transformers, the nominal power of their bus's load is used. For the lines, the total nominal power of the loads that would be disconnected from the external grid in the event of that line's failure is used. The nuclear plant is assumed to be disconnected in the event of any line failure.

2.3.3 Cost Analysis

The cost analysis is separated into penalty costs due to equipment failure and energy import costs. Annual import cost is calculated by classifying all 24 hours into three fee zones:

- **Valley:** 0–8,
- **Flat:** 8–10, 14–18, 22–24,
- **Peak:** 10–14, 18–22,

with corresponding energy price zones:

$$P_{\text{valley}} = 45 \text{ €/MWh}, \quad P_{\text{flat}} = 65 \text{ €/MWh}, \quad P_{\text{peak}} = 90 \text{ €/MWh}.$$

The daily imported energy in each zone is extrapolated to annual values (365 days) to obtain the annual import cost. In addition, a penalty is applied to the calculated unserved energy due to equipment failures using a penalty factor of 90€/MWh, allowing computation of annual failure costs per asset and the total reliability impact.

3 Base Case Results and Diagnosis

This chapter presents the main results obtained from the 24-hour time-series AC power-flow simulation of the current system and provides a diagnosis of the critical technical and economic issues that motivate grid reinforcement in the next phases.

3.1 Bus Voltages

Figure 2 shows the hourly evolution of bus voltage magnitudes over the 24-hour period. The base-case results indicate widespread undervoltage in several buses, with the most critical voltage occurring at **Bus 15**, where the minimum voltage reaches **0.8061 p.u.**. The buses identified with undervoltage events are **Bus 2, 5, 6, 13, 14, and 15**. The most severe undervoltage periods coincide with high-demand hours, suggesting that voltage drops are driven by increased loading and network impedance limitations in the current configuration.

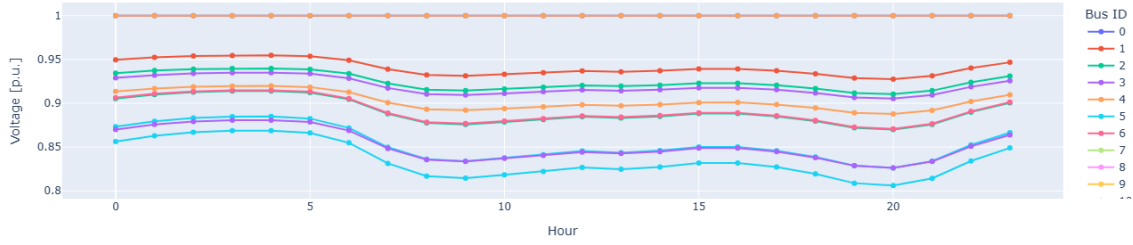


Figure 2: Base Case Bus Voltage Profile

3.2 Demand Coverage

The demand coverage summary for the representative day is reported in Table 1. The total daily demand is **6294.07 MWh**. Local nuclear generation supplies **4198.34 MWh** (about **66.70%** of demand), while the remaining deficit is covered by imports from the external grid (**2353.63 MWh**, about **37.39%** of demand). The fact that generation share plus import share exceeds 100% is explained by network losses, estimated at **257.90 MWh** (about **4.10%** of demand). Figure 3 and Figure 4 illustrate the hourly variations of (i) generation and imports and (ii) individual loads and total demand, respectively. Imports rise notably during high-load hours, confirming that the current system depends strongly on the slack grid to maintain balance.

Table 1: Base-case daily energy balance (representative day).

Quantity	Value
Total demand (MWh)	6294.07
Nuclear generation (MWh)	4198.34
Imports from external grid (MWh)	2353.63
Estimated network losses (MWh)	257.90

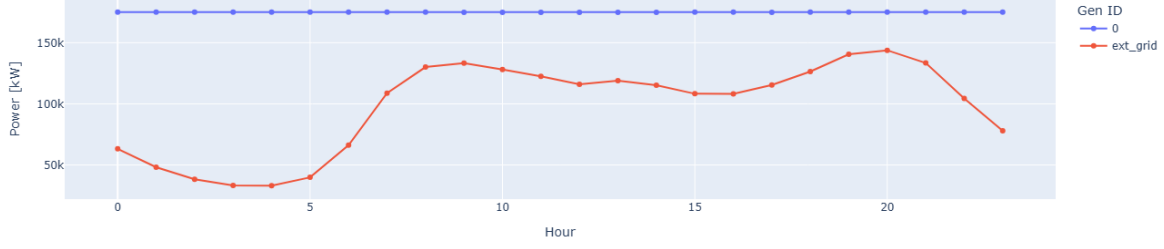


Figure 3: Base Case Generation (Nuclear & Imports)

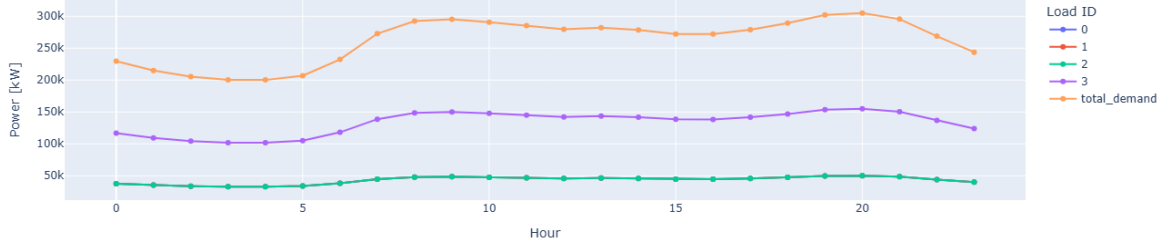


Figure 4: Base Case Load Profiles

3.3 Equipment Loading

The base-case loading results reveal clear thermal constraints in both the line and transformer layers. The worst line loading is observed on **Line 2**, which reaches a maximum loading of **96.97%** and exceeds the 80% threshold for **9 hours**. Transformer loading is more critical: **Transformer 1** reaches a maximum of **119.67%** and exceeds 80% for **20 hours**. The temporal evolution of line and transformer loading is shown in Fig. 5 and Fig. 6. These results indicate that the current network has insufficient capacity in at least one key corridor (Line 2) and one transformer (Transformer 1), which also contributes to the observed undervoltage issues during peak hours.

3.4 Reliability Impact

Using the failure/repair model implemented in 2.3.2, the annualised outage time is estimated as **27.41 hours/year** for lines and **9.55 hours/year** for transformers, resulting in a total of **36.95 hours/year**. The detailed results table indicates that the assets operating at higher loading levels also exhibit higher risk exposure through increased expected downtime. Although the computed annual downtime is moderate in absolute hours, it is operationally relevant for economically sensitive agricultural loads and supports the need for improved redundancy and reduced stress on critical equipment.

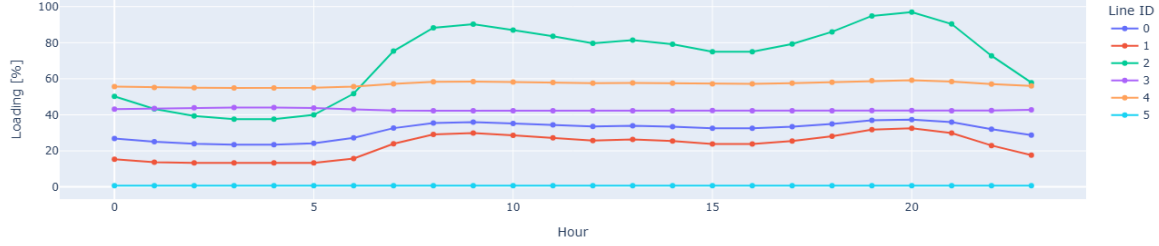


Figure 5: Base Case Line Loading

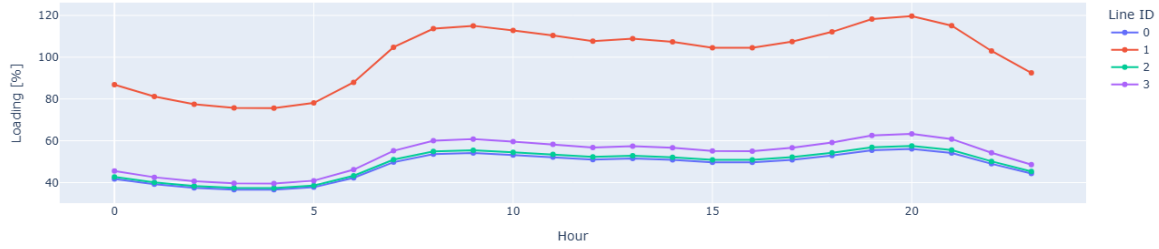


Figure 6: Base Case Transformer Loading

3.5 Annual Operating Cost

The annual operating cost breakdown for the base-case system is presented in Table 2.

Table 2: Annual operating cost breakdown

Cost Component	Amount
Imported energy	€ 62,087,704.40
Transformer failure penalties	€ 72,392.53
Line failure penalties	€ 267,302.08
Total annual cost	€ 62,427,399.01

This represents a total annual expenditure of approximately **62.4 million euros**.

This cost structure indicates that reducing imports (through better network capability and operational flexibility) offers the largest economic leverage, while reliability improvements provide additional benefits through reduced interruption penalties.

3.6 Diagnosis for Grid Upgrades

The base-case results highlight several necessary upgrades for the current system. There are six undervoltage buses at various points of the day, with the most severe being more

than 9% lower than the minimum. This will require additional transmission lines to reduce voltage drops between loads and generation, as well as potentially requiring more central generation to rectify. Extra lines will also help with the overloaded transmission line, and the overloaded transformer will necessarily need a higher capacity as well. Lastly, additional lines and generation will help to increase the network’s efficiency and decrease imported energy, the cost of which is much larger than the reliability penalties.

4 Phase One Upgrades

This chapter presents the Phase 1 upgrade of the Igualada base-case network. The goal of Phase 1 is to improve connectivity and reliability by reinforcing the grid to better satisfy the N-1 criterion, while also ensuring that operational constraints (voltage limits, thermal loading limits, and line-loss requirements) are met. In addition, annual operating cost, failure-related costs, and the required investment cost are computed for the upgraded configuration.

4.1 N-1 Criterion Upgrade

To enhance redundancy and reduce stress on critical corridors, three new single-circuit 110 kV lines were added in the upgraded network. The cost per kilometer figures from [1] are used. These reinforcements create additional supply paths and improve the network’s ability to withstand a single contingency without severe violations. The added lines are summarised in Table 3 and shown in 7.

Table 3: Phase 1 added lines and associated investment cost.

Line name	Length (km)	Voltage (kV)	Circuits	Cost (€)
Vandellos–Falset	21.226	110	1	3,608,458
Lleida–Falset	55.273	110	1	9,396,467
Terrassa–Igualada	33.292	110	1	5,659,678
Total investment				18,664,603

4.2 Voltage Performance

The Phase 1 reinforcement significantly improves voltage conditions relative to the base case. Only **Bus 13** remains below the 0.90 p.u. threshold, with a worst undervoltage of **0.8965 p.u.** and **4 hours** of undervoltage over the day. All other buses remain within acceptable bounds for the full 24-hour simulation. This result indicates that the reinforcements reduce voltage drops across the network, but a small residual undervoltage persists at a critical node, motivating additional corrective measures in the next phase.

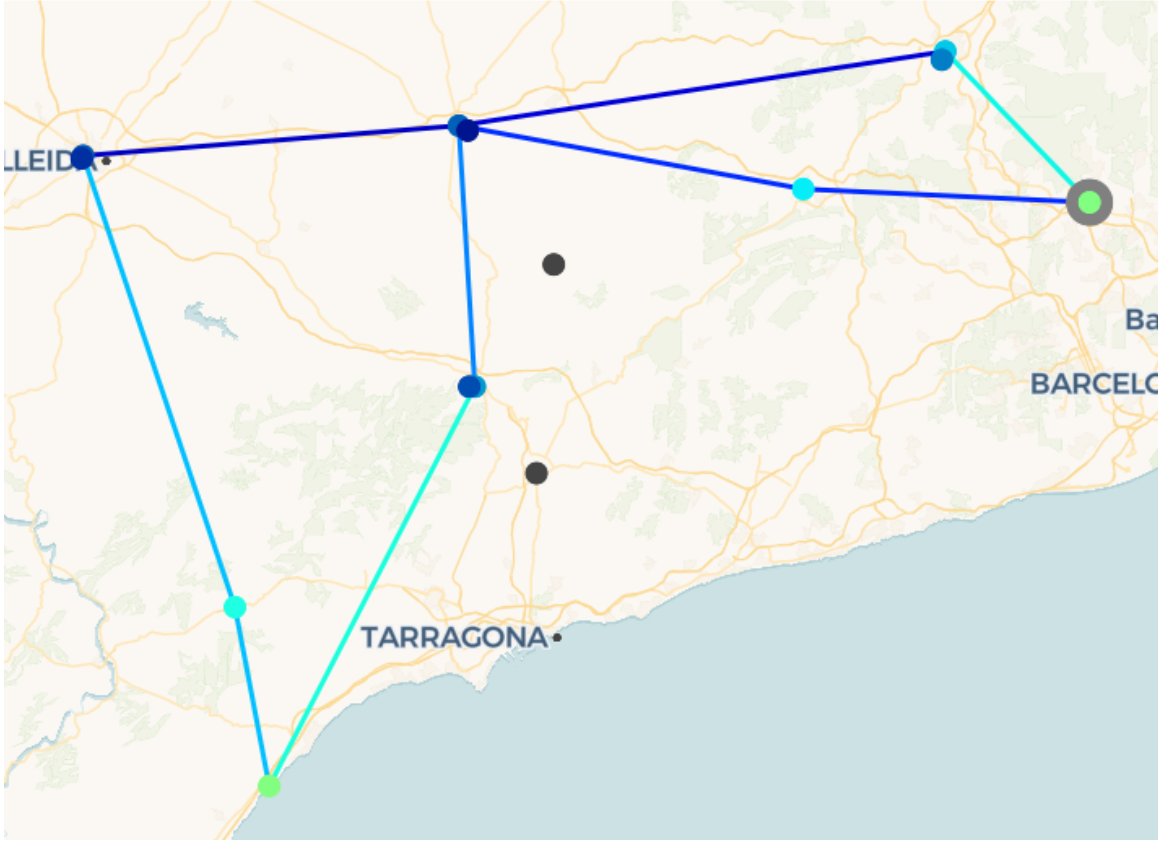


Figure 7: Updated Region Map with N-1 Criterion

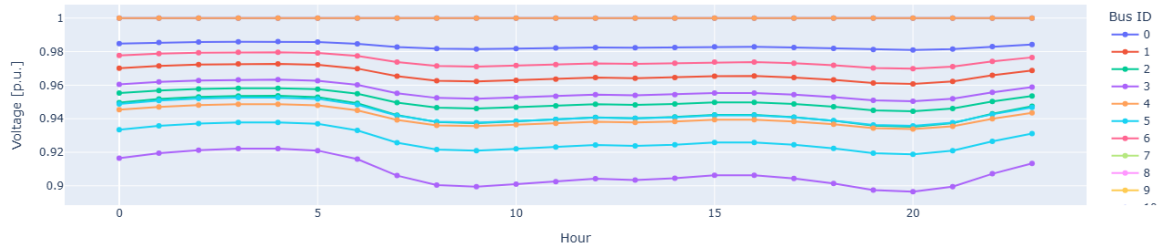


Figure 8: Phase 1 Bus Voltage Profile

4.3 Demand Coverage & Imports

Table 4 summarises the daily energy balance of the upgraded system. Total demand remains **6294.07 MWh**, while nuclear generation supplies **4198.34 MWh (66.70%)**. Imported energy from the external grid decreases to **2241.66 MWh (35.62%)**, showing an improved capability of the network to serve demand with lower dependence on imports. Estimated network losses also reduce to **145.93 MWh (2.32% of demand)**,

representing a substantial reduction compared with the base case.

Table 4: Phase 1 daily energy balance (representative day).

Quantity	Value
Total demand (MWh)	6294.07
Nuclear generation (MWh)	4198.34
Imported energy (MWh)	2241.66
Exported energy (MWh)	0.00
Estimated network losses (MWh)	145.93

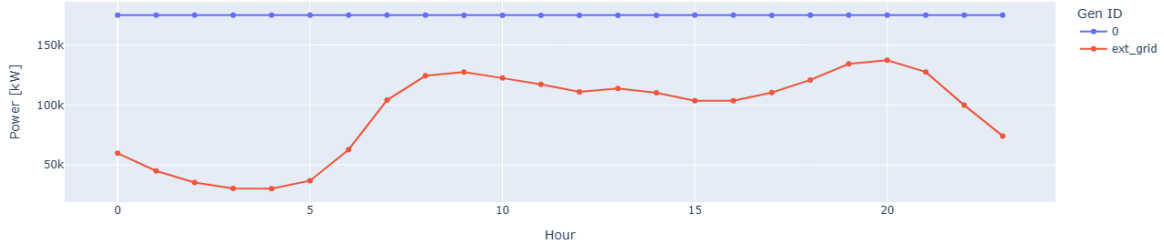


Figure 9: Phase 1 Generation (Nuclear & Imports)

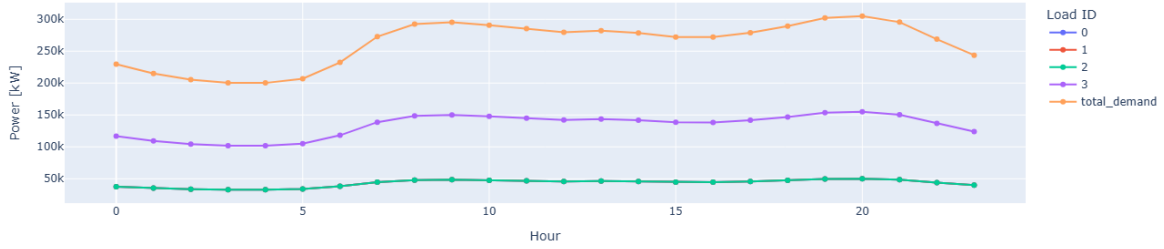


Figure 10: Phase 1 Load Profiles

4.4 Equipment Loading

The Phase 1 reinforcements remove line overload issues: the worst line loading is **59.51%** (Line 2), and there are **0 hours** with any line loading above 80%. However, transformer loading remains the dominant bottleneck. **Transformer 1** reaches a maximum loading of **110.31%** and exceeds 80% for **19 hours**. Therefore, while the added lines successfully increase corridor capacity and reduce line stress, the system

still requires transformer-focused reinforcement (e.g., transformer uprating or parallel transformer) to achieve full compliance. The rating of the overloaded transformer was increased from 160MVA to 250MVA which yielded a benefit of no overloads as the highest loading appears at hour 20 at 68.87%.

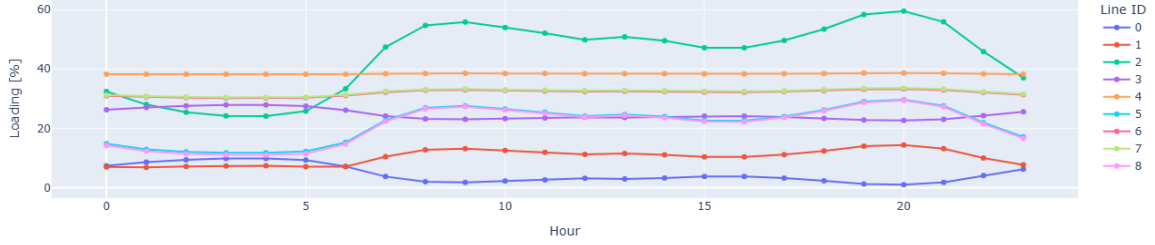


Figure 11: Phase 1 Line Loading

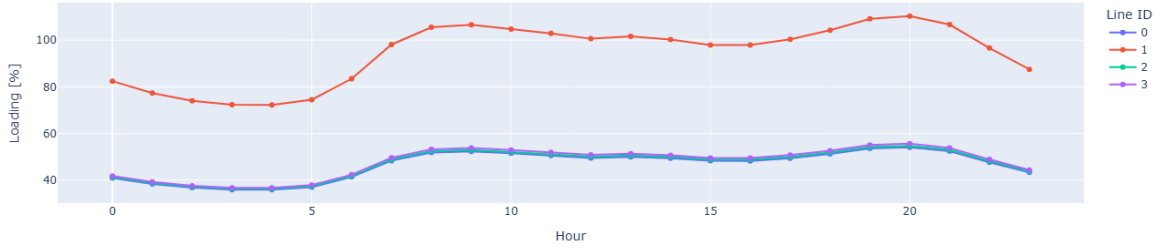


Figure 12: Phase 1 Transformer Loading

4.5 Line Losses

A line-loss compliance check was performed to ensure that the transmission line loss does not exceed the specified 2% requirement. All lines satisfy the constraint (maximum daily loss per line remains well below 2%), and the number of hours with loss above 2% is **zero for all lines**. The computed total annual line losses are **128.21 MWh**. These results confirm that Phase 1 reinforcements meet the line-loss requirement.

4.6 Reliability Costs

Using the implemented failure/repair model, the annual outage time is estimated as **38.44 h/year** for lines and **9.19 h/year** for transformers, yielding a total of **47.63 h/year**. The cost evaluation indicates that the upgraded network achieves **zero transmission line failure cost** under the adopted interruption-cost calculation, while the

Table 5: Loss Analysis Results

Line	Loss Percentage	Hours with Loss greater than 2%
0	0.0359%	0
1	0.1004%	0
2	0.7051%	0
3	0.2015%	0
4	0.6687%	0
5	0.3025%	0
6	0.2028%	0
7	0.5357%	0
8	0.2471%	0

transformer-related failure cost remains non-zero due to the high loading of Transformer 1. This further confirms that the transformer bottleneck is the critical reliability driver in the upgraded system.

4.7 Annual Operating Cost

The annual imported-energy cost under the valley/flat/peak tariff structure is the energy cost. Including failure-related penalty costs, the total annual operating cost of the Phase 1 upgraded system is shown below.

Table 6: Total Annual Operating Cost Breakdown

Cost Component	Amount (€)
Energy cost	59,216,875.18
Transformer Failure cost	69,394.98
Transmission Line Failure cost	0.00
Total Annual Cost	59,286,270.16

Relative to the base case, this represents a notable reduction in annual cost, primarily driven by reduced imports and reduced interruption penalties. Nevertheless, the remaining undervoltage at Bus 13 and the persistent overload of Transformer 1 show that Phase 1 is an improvement but not yet a fully compliant solution.

4.8 Modifying Voltage Level

The current transmission network uses a voltage of 110 kV. However, raising the transmission voltage could have several potential benefits that are studied in this section. A higher voltage level increases the efficiency of transmission, requiring less imported energy and decreasing transformer and line loading. It also requires paying for new line and transformer installations. To determine the savings in voltage losses, the voltage

level of the Phase 1 simulation is raised to 132 kV and 220 kV. The cost analysis using the same method as Section 2.3 for the two different voltage increases are shown in Table 7.

Table 7: Operating Costs of Voltage Level Increase

Cost Component	Phase 1 Base (€)	132 kV (€)	220 kV (€)
Energy Import	59,216,875.18	58,153,188.94	56,805,398.59
Transformer Failure	69,394.98	67,067.58	64,376.51
Line Failure	0.00	0.00	0.00
Total Annual Cost	59,286,270.16	58,220,256.52	56,869,775.1
Annual Savings		1,066,013.64	2,416,495.06

It is clear that the increased efficiency significantly reduced the cost of energy imported from the external grid, and the decreased loading on the HV side of the transformer also reduced failure costs. Next, the cost of the equipment is considered to determine the long-term economic feasibility of upgrading the network voltage. [1] gives unit investment costs for electrical equipment. The mean cost for 150-280 MVA transformers is used for the Tarrega transformer, and the mean cost per MVA is used for the three smaller transformers. The smallest transmission line presented is the cost per km for 220 kV single and double circuit lines. The mean cost is used for 220 kV transmission lines, and the minimum inter-quartile value is assumed for the 132 kV upgrade. The results are shown in Table 9.

Table 8: Transmission Line Costs

Line Type	Cost/km (€)
220 kV Single	288,289
220 kV Double	407,521
132 kV Single	157,926
132 kV Double	354,696

Table 9: Equipment Costs of Voltage Level Increase

Cost Component	132 kV (€)	220 kV (€)
Transformers	4,526,407.00	4,526,407.00
Lines	83,891,803.38	119,546,705.34
Total	88,418,210.38	124,073,112.34

According to these results, it would take 50.48 years for the 220 kV upgrade to pay for itself and 82.94 years for the 132 kV upgrade. Both of these are beyond the typical limit for a transformers' useful life of 20-30 years [2], so it is therefore not likely that a voltage increase would have any benefits. This estimation is based on several

assumptions about the equipment pricing and simulated behavior. The transformer parameters such as core losses and short circuit voltage are assumed to be unchanged from the base case. Additionally, [1] takes the transformer costs to be solely proportional to the MVA rating, so the transformer costs are the same between the two cases even though the 220 kV transformers would likely be more expensive.

4.9 Phase One Diagnosis

Phase 1 achieves several key improvements. The additional lines satisfy the N-1 criterion, which reduces the line failure costs to zero, as all loads can be supplied in the event of any single line failure. The additional lines also provide more efficient transmission, reducing losses and import costs and bringing the maximum line loading below 80% for all lines at all hours. The additional transmission flexibility also rectifies the majority of the undervoltage cases by reducing voltage drops across highly loaded transmission lines. Lastly, the upgraded transformer eliminates the transformer overloading at Tàrraga LV (Bus 13) and eliminates the last undervoltage case there at hour 20.

Phase 2 upgrades will focus on decreasing energy import costs, which still remain quite high due to the lack of generation in the base case and upgraded network.

5 Phase Two Upgrades

The next phase of upgrades will consider the implementation of various modern power grid technologies to further reduce the cost of imported energy. The explored solutions are renewable energy sources, battery energy storage connected by HVDC, and demand response programming. Each solution is implemented and simulated separately from the Phase 1 base case, and the effects on system operation and an economic estimation are made for each.

5.1 Solution 1: RES Integration

This section presents the Phase 2 upgrade (Solution 1), where renewable energy sources (RES) are integrated into the upgraded network to:

- (i) Reduce the dependency on imported energy,
- (ii) Improve voltage conditions (especially the residual undervoltage at Bus 13 identified after Phase 1), and
- (iii) Maintain compliance with operational constraints (thermal loading and transmission-line loss requirement).

As in previous stages, the assessment is based on a 24-hour time-series AC power-flow simulation and an annualized cost and reliability evaluation.

5.2 Phase 2 modifications: N-1 reinforcement and RES additions

5.2.1 Additional lines to meet N-1 criterion

To improve connectivity and redundancy in Phase 2, two additional 110 kV single-circuit lines are introduced to strengthen the Igualada zone and provide alternative supply paths. Table 10 summarises the new corridors and their investment costs.

Table 10: Phase 2 (Solution 1) added lines and associated investment cost.

Line name	Length (km)	Voltage (kV)	Circuits	Cost (€)
Tàrrrega–Conesa	19.48	110	1	3,312,332.90
Igualada–Conesa	30.28	110	1	5,148,286.91
Total Phase 2 line investment				8,460,619.81

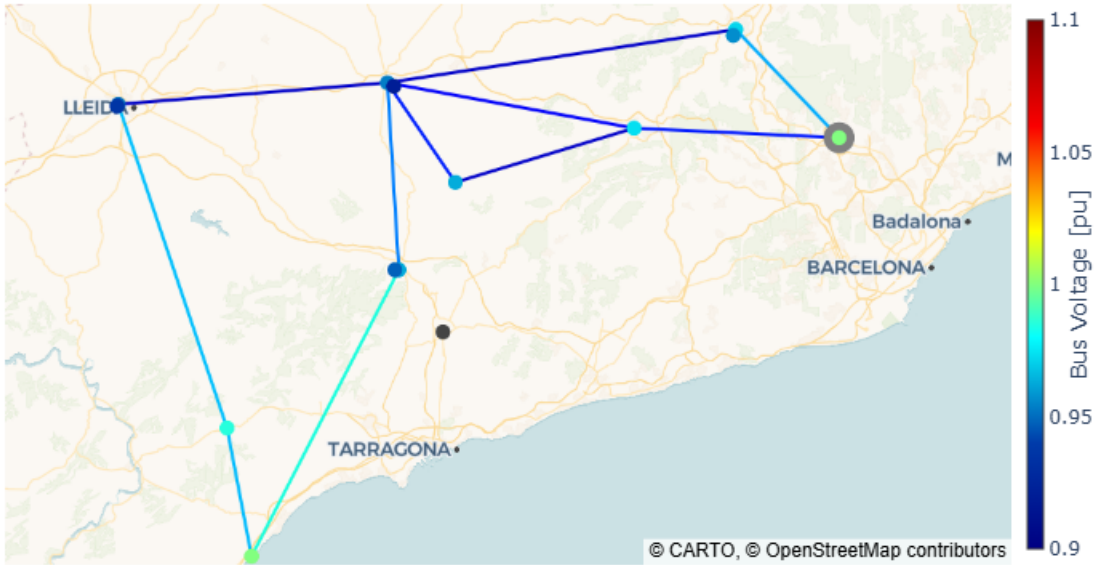


Figure 13: Updated Region Map with N-1 Criterion

5.3 Derivation of Installed RES Capacities (Falset PV and Conesa Wind)

This section explains how the installed capacities for Falset (PV) and Conesa (wind) were derived, following the sizing logic implemented in the project code.

5.3.1 Falset Solar PV Capacity

Falset is allocated a land area of $A_{\text{ha}} = 150$ ha. Using the conversion $1 \text{ ha} = 10,000 \text{ m}^2$, the total land area in square meters is

$$A = 150 \times 10,000 = 1,500,000 \text{ m}^2. \quad (1)$$

The PV ground power density (in Wp/m^2 of land) is computed from the module efficiency $\eta = 0.20$, the ground coverage ratio $\text{GCR} = 0.40$ [3], and the peak irradiance $G = 1000 \text{ W/m}^2$:

$$p_d = \eta \cdot \text{GCR} \cdot G = 0.20 \times 0.40 \times 1000 = 80 \text{ Wp/m}^2. \quad (2)$$

Therefore, the installed peak PV capacity is

$$P_{\text{Falset}} = \frac{A \cdot p_d}{10^6} = \frac{1,500,000 \times 80}{10^6} = 120 \text{ MWp}. \quad (3)$$

5.3.2 Conesa Wind Capacity

Conesa is allocated a land area of $A_{\text{ha}} = 550$ ha, which corresponds to

$$A_{\text{km}^2} = \frac{550 \times 10,000}{10^6} = 5.5 \text{ km}^2. \quad (4)$$

A Vestas V162 turbine is assumed, with rotor diameter $D = 162 \text{ m} = 0.162 \text{ km}$ and rated power $P_r = 6.2 \text{ MW}$ [6]. The spacing rule is $7D$ along the prevailing wind direction and $5D$ in the cross-wind direction, giving

$$s_{\parallel} = 7D = 7 \times 0.162 = 1.134 \text{ km}, \quad s_{\perp} = 5D = 5 \times 0.162 = 0.81 \text{ km}. \quad (5)$$

The land “footprint” per turbine is then approximated as a rectangular area:

$$A_{\text{turb}} = s_{\parallel} s_{\perp} = 1.134 \times 0.81 = 0.91854 \text{ km}^2. \quad (6)$$

Following the implementation in the code, the maximum number of turbines is computed using the total available area:

$$N = \left\lfloor \frac{A_{\text{km}^2}}{A_{\text{turb}}} \right\rfloor = \left\lfloor \frac{5.5}{0.91854} \right\rfloor = 5. \quad (7)$$

Finally, the installed wind capacity is

$$P_{\text{Conesa}} = N \cdot P_r = 5 \times 6.2 = 31 \text{ MW}. \quad (8)$$

5.3.3 RES integration and motivation (mitigating Bus 13 undervoltage)

Following Phase 1, the remaining critical issue was a residual undervoltage at Bus 13. To address this, additional local RES generation was introduced (notably around the Conesa node and other RES sites), which improved the voltage support in the affected corridor. With the RES integration, the minimum voltage at **Bus 13 improves to approximately 0.899 p.u. and the undervoltage duration is reduced to only one hour. In practice, rounding to one decimal place yields 0.90 p.u.**, meaning the residual undervoltage is effectively eliminated under standard reporting precision.

5.4 Time-series simulation and recorded outputs

The upgraded network (Phase 2, Solution 1) is evaluated using the same 24-hour representative-day profiles and the `pandapower.timeseries` workflow. Hourly results are recorded for:

1. Bus voltages (V in p.u.),
2. Line/transformer loading (%),
3. Generation, imports/exports at the external grid (MW),
4. Transmission line loss indicators (%), and
5. Annualized operating cost and failure-related penalties.

5.5 Results and diagnosis (Phase 2, Solution 1)

5.5.1 Voltage profile and deviation assessment

The voltage deviation screening identifies only **Bus 13** as marginally critical. The minimum voltage occurs at Bus 13 with

$$V_{\min,13} \approx 0.899 \text{ p.u.}, \quad \text{Hours under 0.90 p.u.} = 1 \text{ hour.}$$

All other buses remain within acceptable bounds during the full 24-hour cycle.

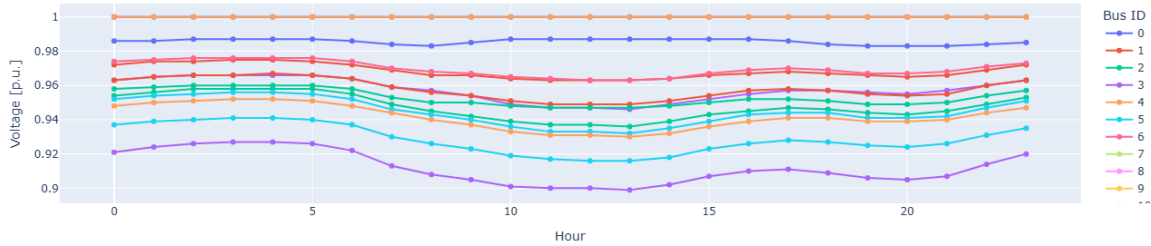


Figure 14: Phase 2 (Solution 1) bus voltage profile over 24 hours

5.5.2 Demand coverage, imports/exports, and energy balance

Table 11 summarises the daily energy balance for the representative day. **Total demand remains 6294.07 MWh.** With RES integration, solar and wind contribute a significant share of the daily energy, thereby reducing **imported energy to 1033.84 MWh.** A small amount of **export (92.98 MWh)** is also observed, indicating that RES output exceeds local consumption at certain hours. **Network losses are estimated at 175.96 MWh** (about 2.8% of demand).

Table 11: Phase 2 (Solution 1) daily energy balance (representative day).

Quantity	Value
Total demand (MWh)	6294.07
Nuclear generation (MWh)	4198.34
Solar generation (MWh)	803.74
Wind generation (MWh)	527.09
Total generation (MWh)	5529.17
Energy imported from grid (MWh)	1033.84
Energy exported to grid (MWh)	92.98
Network losses (est.) (MWh)	175.96

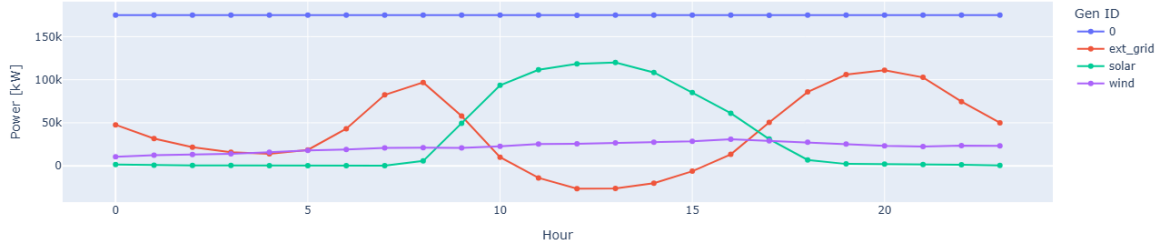


Figure 15: Phase 2 (Solution 1) hourly generation and external-grid power exchange

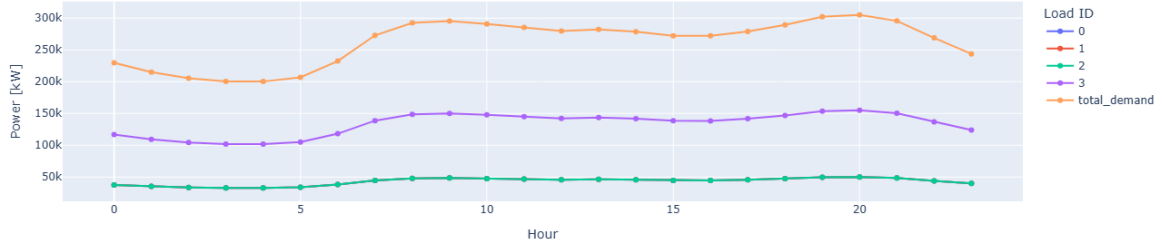


Figure 16: Phase 2 hourly load profiles and total demand

5.5.3 Equipment loading (lines and transformers)

Thermal loading results show no line overloads in Phase 2 (Solution 1). The **worst line loading is 60.33%** (Line 4), with zero hours exceeding the **80%** threshold. However, the transformer bottleneck remains: Transformer 1 reaches a **maximum loading of 109.32%** and exceeds **80%** for **19 hours**. This confirms that line reinforcements and RES integration improve corridor utilization and voltage support, but do not fully resolve the dominant transformer capacity constraint.

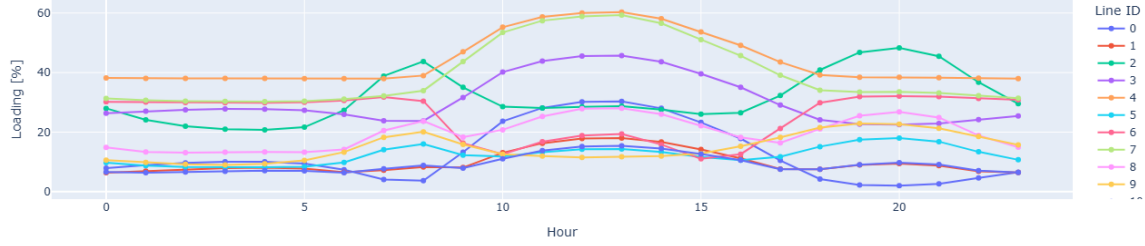


Figure 17: Phase 2 (Solution 1) line loading over 24 hours

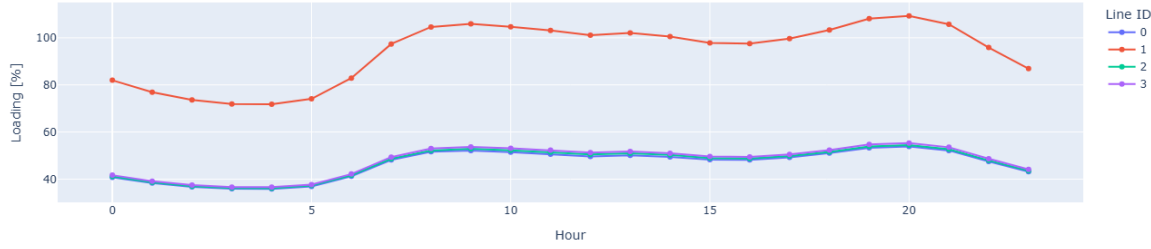


Figure 18: Phase 2 (Solution 1) transformer loading over 24 hours

5.5.4 Transmission line loss requirement ($< 2\%$)

The loss compliance check confirms that all transmission lines satisfy the requirement $\text{loss} < 2\%$, with zero hours above **2%** for every line. The maximum per-line losses remain below **1.7%** in the simulation, and the total annual line losses are estimated as **158.31 MWh**. Therefore, Phase 2 (Solution 1) satisfies the line-loss constraint.

5.5.5 Annual operating cost and failure-related penalties

The annual imported-energy cost (valley/flat/peak tariff framework) is:

BESS significantly reduces operating costs by lowering imported energy at peak times and shifting it to cheaper hours, while also allowing exports. The remaining penalty cost is primarily driven by the persistent overload of Transformer 1.

This confirms that RES integration substantially reduces operating cost by lowering imported energy, while the remaining penalty cost is mainly transformer-driven due to the persistent Transformer 1 overload.

5.6 Investment cost of RES (solar and wind plants)

For the RES investment estimation in Solution 1, Table 14 summarises the CAPEX of the added renewable plants at Falset (solar PV) and Conesa (onshore wind). Based on

Table 12: Loss Analysis Results

Line	Loss Percentage	Hours with Loss greater than 2%
0	0.3495	0
1	0.1565	0
2	0.4638	0
3	0.5435	0
4	1.6403	0
5	0.1103	0
6	0.1887	0
7	1.6862	0
8	0.2227	0
9	0.0882	0
10	0.0604	0

Table 13: Annual Operating Cost Breakdown

Cost Component	Annual Cost (€/year)
Energy Cost	23,498,049.91
Transformer Failure	69,112.72
Line Failure	0.00
Total Annual Operating Cost	23,567,162.63

the sizing results, the Falset solar PV plant has an installed capacity of 120 MWp and a unit investment cost of 588.34 €/kW, in addition, the Conesa onshore wind plant is sized at 31 MW with a unit investment cost of 886.33 €/kW[4].

Table 14: Investment cost of RES plants (Solution 1).

Location	Capacity (MW)	CAPEX (€/kW)	CAPEX Total (€)
Falset	120.00	588.34	70,600,255.43
Conesa	31.00	886.33	27,476,372.92

These RES investment costs represent the generation-side CAPEX component of Solution 1 and should be assessed alongside the Phase 2 line reinforcement investments and the associated reductions in annual operating (import) costs.

5.7 Summary of findings (Solution 1 diagnosis)

Phase 2 (Solution 1) achieves the following:

- **Reduced import dependence:** imported energy decreases markedly (to about 1033.84 MWh for the representative day), leading to a much lower annual energy cost.

Table 15: Annual cost comparison for base case, Phase 1, and Phase 2 (Solution 1).

Cost Component	Base Case (€)	Phase 1 (€)	Solution 1 (€)
Energy Import Cost	62,087,704.40	59,216,875.18	23,498,049.91
Trafo Failure Penalty	72,392.53	69,394.98	69,112.72
Line Failure Penalty	267,302.08	0.00	0.00
Total Annual Cost	62,427,399.01	59,286,270.16	23,567,162.63
<i>Savings vs Base</i>	-	3,141,128.85	38,860,236.38

- **Voltage improvement:** the residual undervoltage at Bus 13 is reduced to only one hour with $V_{\min} \approx 0.899$ p.u., effectively meeting the limit under standard rounding/reporting.
- **Line thermal compliance and loss compliance:** no line exceeds 80% loading and all lines satisfy the $< 2\%$ loss requirement.

However, the **transformer bottleneck remains**, with Transformer 1 still exceeding its thermal rating for most of the day (max 109.32%, 19 hours $> 80\%$). Therefore, while RES integration resolves the major economic issue (imports) and substantially improves voltage performance, additional measures in subsequent solutions (e.g., transformer reinforcement, alternative power-routing strategies, or storage/HVDC options) are still required to achieve full thermal compliance and improved reliability margins.

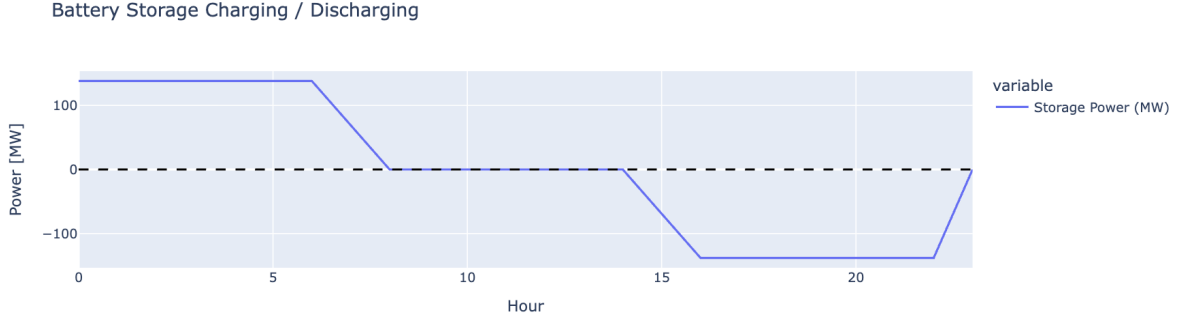


Figure 20: BESS Storage System Charge and Discharge Profile

from the cost effectiveness of HVDC transmission, which requires only two conductors compared to three in a conventional AC circuit.

5.9 Results and diagnosis (Phase 2, Solution 2)

5.9.1 Voltage profile and deviation assessment

The voltage deviation screening identifies only **Bus 13** as marginally critical. The minimum voltage occurs at Bus 13 with

$$V_{\min,13} \approx 0.897 \text{ p.u.}, \quad \text{Hours under } 0.90 \text{ p.u.} = 3 \text{ hours (hours 19, 20 and 21)}.$$

As illustrated in Fig. 21, all other buses remain within acceptable voltage limits throughout the full 24-hour cycle.

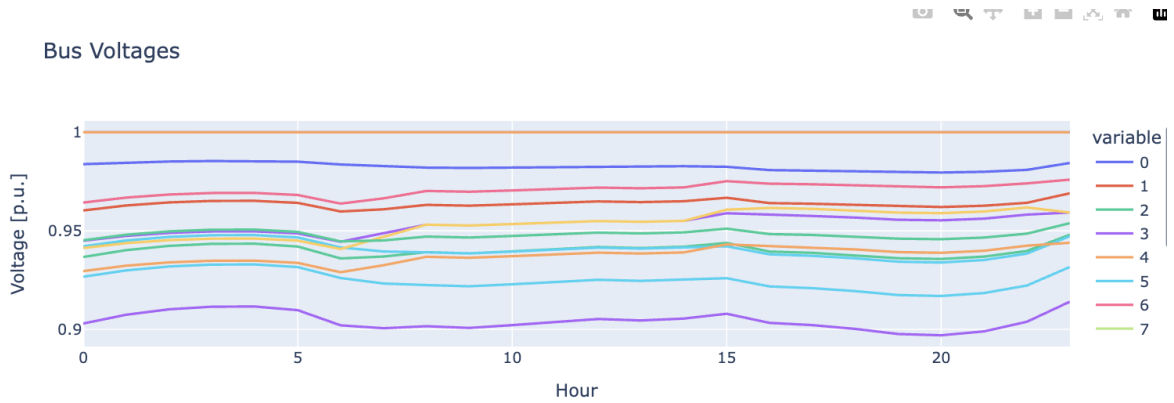


Figure 21: Bus voltages with BESS and HVDC integration over 24 hours

5.9.2 Demand coverage, imports/exports, and energy balance

Table 16 summarises the daily energy balance for the representative operating day in Phase 2 (Solution 2), including the operation of the battery energy storage system

(BESS). The total system demand amounts to 7329.07 MWh, which includes both consumer loads and the additional electricity required for BESS charging during off-peak hours. This increase does not correspond to a net rise in energy consumption, as the stored energy is subsequently discharged during the evening peak period.

The BESS charging demand totals 1035.00 MWh over the day. Nuclear generation contributes 4198.34 MWh, forming the dominant local generation source, while total generation reaches 5223.34 MWh when accounting for all internal sources. The remaining demand is supplied by imports from the external grid, amounting to 2403.70 MWh.

During periods of low demand combined with high nuclear output and BESS discharge, surplus energy is exported to the external grid. These exports total 114.10 MWh over the day. Network losses are estimated at 73.87 MWh, corresponding to approximately 1.0% of total system demand, indicating efficient overall network operation.

Table 16: Phase 2 (Solution 2) daily energy balance (representative day) including BESS.

Quantity	Value (MWh)
Total demand (loads + BESS)	7329.07
BESS charging demand	1035.00
Nuclear generation	4198.34
Total generation	5223.34
Energy imported from grid	2403.70
Energy exported to grid	114.10
Network losses (estimated)	73.87

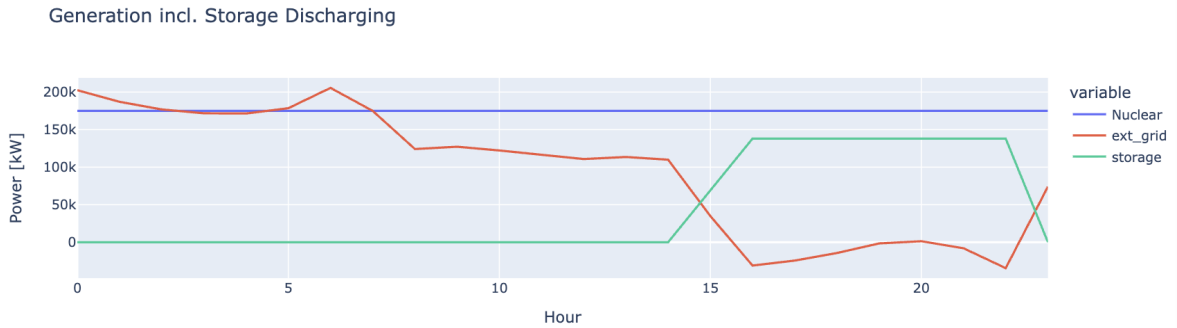


Figure 22: Generation profile across the 24 hour period

5.9.3 Equipment loading (lines and transformers)

Thermal loading results show no line overloads in Phase 2 (Solution 2). The worst line loading reaches **78.9%** on Line 4, with no hours exceeding the **80%** loading threshold, as shown in Fig. 23. However, the transformer bottleneck remains evident. Transformer 1 reaches a maximum loading of **110.25%** and exceeds the **80%** threshold for

19 hours over the day, as illustrated in Fig. 24. These results confirm that while line reinforcements and energy storage integration improve corridor utilisation and voltage support, they do not fully resolve the dominant transformer capacity constraint.

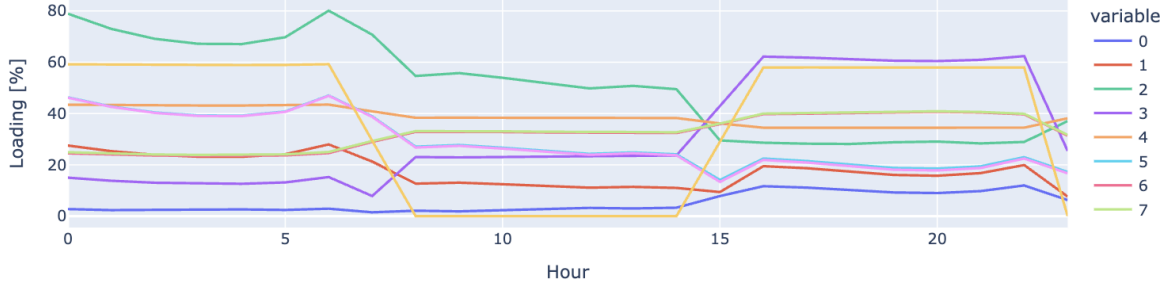


Figure 23: Line loading after BESS integration over 24 hours

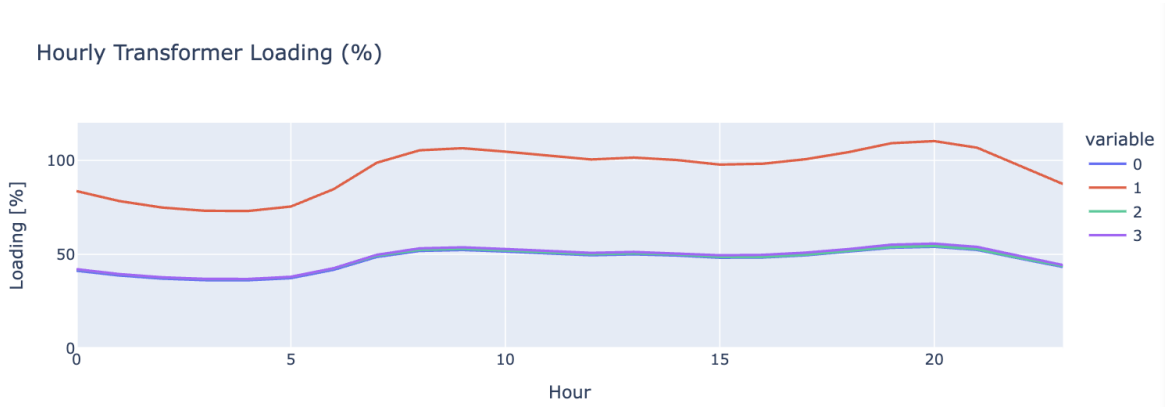


Figure 24: Transformer loading after BESS integration over 24 hours

5.9.4 Transmission line loss requirement ($< 2\%$)

The loss compliance check confirms that all transmission lines satisfy the requirement $\text{loss} < 2\%$, with zero hours above **2%** for every line. The maximum per-line losses remain below **1.28%** in the simulation, and the total annual line losses are estimated as **78.87 MWh**. Therefore, Phase 2 (Solution 2) satisfies the line-loss constraint.

Table 17: Loss Analysis Results

Line	Loss Percentage	Hours with Loss greater than 2%
0	0.055	0
1	0.394	0
2	1.281	0
3	1.013	0
4	0.853	0
5	0.384	0
6	0.301	0
7	0.798	0
8	0.629	0
9	0.219	0

5.9.5 Energy Balance and Annual Costs

The annual operating costs of the system, including energy procurement and network reliability impacts, are summarised in Table 18. Energy procurement constitutes the dominant cost component, while line failure costs are fully mitigated in this solution. Transformer-related costs remain non-zero due to persistent loading constraints, resulting in a total annual operating cost of **€48.35 million**.

Table 18: Annual operating costs for the BESS and HVDC connection

Cost Component	Annual Cost (€ / year)
Energy cost	48,268,299.81
Transformer failure	77,710.64
Line failure	0.00
Total annual operating cost	48,346,010.45

5.9.6 CAPEX for Energy Storage and HVDC Connection

The capital expenditure (CAPEX) associated with the BESS installation and HVDC connection infrastructure is detailed in Table 20. The BESS storage system represents the largest share of the investment, while the HVDC transmission line and converter stations account for a comparatively smaller portion of the total cost. The overall CAPEX for the BESS–HVDC connection amounts to **€352.25 million**.

5.10 Investment cost of BESS and HVDC connection

For the investment estimation in Solution 2, Table 21 summarizes the CAPEX of connecting the Valls BESS via HVDC and the associated VSC converters. The Valls BESS plant has an installed capacity of 138 MW / 966 MWh, with a unit investment cost of

Table 19: CAPEX for Energy Storage and HVDC Connection Components

Component	Capacity / Size	Unit Cost (€)	Total CAPEX (€)
Valls BESS (Storage)	966 MWh / 138 MW	350 €/kWh	338,100,000
VSC Converter Stations	135 MVA	87,173 €/MVA	11,768,355
HVDC Transmission Line	12.35 km	288,289 €/km $\times$ 0.67 factor	2,385,447
Total BESS connection CAPEX	—	—	352,253,802

Table 20: CAPEX for Energy Storage and HVDC Connection Components

350 €/kWh, resulting in a total investment of 338.1 million €. The VSC converter stations required for the connection cost 11.77 million €, while the HVDC transmission line costs 2.39 million €, bringing the total BESS connection investment to approximately 352.26 million €.

Table 21: Investment cost of BESS and HVDC connection plants (Solution 2).

Component	Capacity / Size	Unit Cost (€)	Total CAPEX (€)
Valls BESS (Storage)	966 MWh / 138 MW	350 €/kWh	338,100,000
VSC Converter Stations	135 MVA	87,173 €/MVA	11,768,355
HVDC Transmission Line	12.35 km	288,289 €/km $\times$ 0.67 factor	2,385,447
Total BESS Connection CAPEX	—	—	352,253,802

These BESS and HVDC investments represent the Storage and HVDC CAPEX component of Solution 2 and should be assessed alongside Phase 2 line reinforcement investments and the associated reductions in annual operating costs.

5.11 Summary of Findings (Solution 2 Diagnosis)

Phase 2 (Solution 2) achieves the following:

- **Optimised importing:** Net imported energy amounts to 2,289.3 MWh, but at lower costs compared to the no storage scenario in Phase. This is achieved by charging the BESS during cheaper night-time hours and discharging during peak evening hours, thus avoiding more expensive imports.
- **Voltage improvement:** The residual undervoltage at Bus 13 occurs for three hours, with $V_{\min} \approx 0.897$ p.u., effectively meeting the standard limit under rounding and reporting conventions.
- **Line thermal and loss compliance:** No transmission line exceeds 80% loading, and all lines satisfy the less than 2% loss requirement.

Table 22 summarises the annual cost comparison for the Base Case, Phase 1, Phase 2 (Solution 1) and Phase 2 (Solution 2):

However, the transformer bottleneck remains. Transformer 1 continues to exceed its thermal rating for most of the day, with a maximum loading of 110.26% and 19 hours

Table 22: Annual cost comparison for Base Case, Phase 1, Phase 2 (Solution 1) and Phase 2 (Solution 2)

Cost Component	Base Case (€)	Phase 1 (€)	Solution 1 (€)	Solution 2 (€)
Energy import cost	62,087,704	59,216,875	23,498,049.91	48,268,299.81
Transformer failure penalty	72,392	69,394	69,112.72	77,716.84
Line failure penalty	267,302	0	0.00	0.00
Total annual cost	62,427,399	59,286,270	23,567,162.63	48,346,016.65
Savings vs Base Case	–	3,141,129	38,869,936.37	14,091,082.35

above 80%. While integrating the BESS via HVDC improves voltage performance, this comes at a significant € 352M economic investment compared to the actual savings €13M and the typical 20 year lifespan of BESS technology. In addition, further measures including uprating the overloaded transformer, are required in subsequent solutions to fully meet voltage and thermal requirements in the network.

5.12 Solution 3: Demand Response

The third solution proposed for the Phase 2 upgrade involves the implementation of a **Demand Response (DR)** program. Unlike previous solutions that focused on physical infrastructure reinforcement (Phase 1) or the integration of renewable generation, this approach focuses on the demand side of the grid.

The primary objective of Solution 3 is to optimize the system’s operational efficiency by modifying the consumption profiles of the loads. By shifting energy consumption from high-cost, high-demand “Peak” hours to lower-cost “Valley” hours, the grid operator aims to:

1. Reduce the peak thermal stress on critical equipment (transformers and lines).
2. Minimize the annual operating costs by arbitraging the electricity price difference between time zones.
3. Flatten the daily load curve to improve voltage stability and demand coverage.

This section evaluates the technical and economic impact of applying a load-shifting mechanism to the Phase 1 upgraded network.

5.12.1 Methodology: Load Shifting Strategy and Constraints

The Demand Response mechanism was modeled by applying a load shifting algorithm to the hourly demand profiles of the cities (Lleida, Tàrraga, Manresa, and Montblanc). The strategy is defined by the electricity tariff periods established in the project instructions:

- **Valley (Off-Peak):** 0, 1, 2, 3, 4, 5, 6, 7 (Low price)

- **Flat (Standard):** 8, 9, 14, 15, 16, 17, 22, 23 (Medium price)
- **Peak (High-Demand):** 10, 11, 12, 13, 18, 19, 20, 21 (High price)

The simulation applies a percentage-based shift where a portion of the load during **Peak** hours is curtailed and moved to **Valley** hours. This simulates behavioral changes in consumers triggered by price signals.

Important note on Tariff-Profile Mismatch: It is important to note a specific structural discrepancy observed during the modeling of this strategy. The “Valley” tariff period (low price) is defined as ending at 07:00. However, the specific load profiles for the region show a natural morning ramp-up beginning prior to 07:00.

Consequently, the DR algorithm optimizing purely for cost shifts load into the final hours of the Valley period (06:00–07:00). This shifted load coincides with the natural morning rise in demand, creating a higher peak demand where the total power (Base Load + Shifted Load) exceeds the original peak demand. This phenomenon influences the thermal loading results discussed below.

5.12.2 Simulation Results

Voltage Performance The implementation of Demand Response resulted in a voltage profile that largely mirrors the Phase 1 baseline but with slight deviations during the new peak periods created by load shifting.

- **Maximum Voltage:** The system maintained a maximum voltage of **1.00 p.u.** at the generation buses.
- **Minimum Voltage:** The lowest recorded voltage dropped slightly to **0.893 p.u.**, occurring at Bus 13. This represents a marginal degradation compared to Phase 1.

While the voltage profile remains largely stable, the residual undervoltage at Bus 13 persists. The load shift did not sufficiently alleviate the voltage drop at the end of the radial feeder.

Thermal Loading and Equipment Stress The impact of Demand Response on equipment loading was mixed, confirming the increase in peak demand described in the methodology.

Transmission Lines: The transmission lines remain well within their thermal limits. The maximum line loading recorded was **63.99%** (Line 2: Terrassa–Manresa). This is a slight increase from Phase 1 (59.49%) but remains safely below the 80% safety threshold.

Transformers (The Impact of the Rebound Peak): Crucially, Solution 3 failed to resolve the critical bottleneck at **Transformer 1 (Tàrrega)** and, in fact, worsened the peak loading.

- **Trafo 1 Max Loading: 114.79%.**
- **Comparison:** This is an *increase* compared to the Phase 1 result of 110.31%.

This result is a direct consequence of the mismatch between the tariff definition and the load profile. By pushing peak usage into the “Valley” slot (specifically the 07:00–08:00 window), the algorithm created a new, higher aggregate peak. This indicates that a simple price-based shift without a “maximum peak constraint” or a redefinition of the tariff hours is dangerous for transformer health.

Demand Coverage and Generation vs. Load To visualize the impact of the solution, the generation and load profiles of all three scenarios were analyzed. Figure 25 presents the evolution from the Base Case to Phase 1 and finally Solution 3 side-by-side.

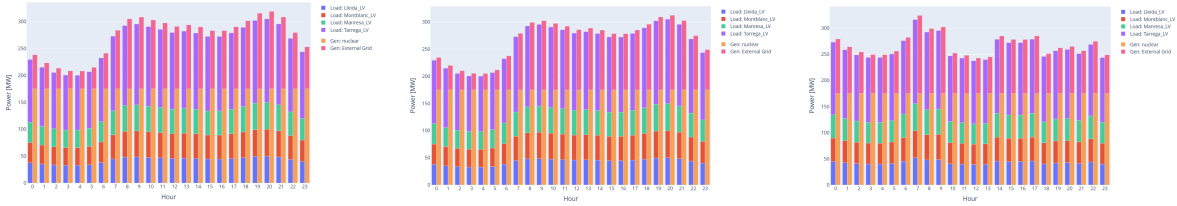


Figure 25: Evolution of Generation vs. Load Profiles. **Left:** Base Case. **Center:** Phase 1. **Right:** Solution 3 (Demand Response).

As seen in Figure 25 (Right), the DR implementation successfully flattens the import curve during the midday peak (10:00–14:00) compared to the Base Case and Phase 1. However, the “Valley Filling” creates a sharp spike immediately before 08:00, confirming the source of the transformer overload. In all cases, the generation area slightly exceeds the load line, accounting for active power losses ($< 2\%$).

Annual Operating Costs The most significant positive impact of Solution 3 is economic. Table 15 details the cost breakdown.

Table 23: Annual Cost Comparison for Base Case, Phase 1, and Phase 2 Solutions

Cost Component	Base Case (€)	Phase 1 (€)	Solution 1 (€)	Solution 2 (€)	Solution 3 (€)
Energy Import Cost	62,087,704	59,216,875.18	23,498,049.91	48,268,299.81	53,252,097
Trafo Failure Penalty	72,393	69,394.98	69,112.72	77,716.84	70,854
Line Failure Penalty	267,302	0	0	0	0
Total Annual Cost	62,427,399	59,286,270.16	23,567,162.63	48,346,016.65	53,322,951
Savings vs Base	-	3,141,128.84	38,869,936.37	14,091,082.35	9,104,448

By successfully shifting consumption to cheaper time zones, the system achieved a saving of approximately **6 million €** compared to Phase 1 and **9 million €** compared to the Base Case. However, the reliability costs (approx. 72k €) remain non-zero due to the transformer overload.

5.12.3 Conclusion and Recommendation for Solution 3

Solution 3 (Demand Response) proves to be an economically efficient option for reducing Operational Expenditures (OPEX). The shifting of loads significantly lowers the electricity bill for the region.

However, from a technical perspective, Solution 3 is insufficient as a standalone solution.

1. **Exacerbated Thermal Bottlenecks:** Due to the misalignment between the “Valley” tariff definition and the actual load profile ramp-up, the load shifting created a high load value at hour 7 driving Transformer 1 loading to 114.79%.
2. **Unresolved Voltage Issues:** It fails to correct the undervoltage at Bus 13.

Recommendation: Demand Response should be adopted only as a complementary measure to physical upgrades. Furthermore, the tariff structure definition requires adjustment to match with the actual load profile; the “Valley” period should end earlier to prevent the algorithm from stacking shifted load onto the natural morning ramp. Ideally, Solution 3 should be combined with the physical capacity increases of Solution 1 or 2 to achieve both cost efficiency and N-1 security.

SGAM Layer Design

Name	Agriculture Energy Hub at Igualada
Scope	Igualada is surrounded by farmland and seeks to modernise irrigation systems with renewable-powered pumps and smart meters. Farmers want to schedule water and energy use according to price signals and weather forecasts, reducing operating costs while increasing efficiency.
Objective	Supply low cost electricity and minimise water consumption during the irrigation process in Igualada agriculture
Related Problems	- Inefficient use of electricity and water consumption based on routine irrigation rather than predictive necessity - Dependency on the grid for electricity and thus price fluctuations - No equipment fault or water leak detection
Relation to other Use Cases	High-Level Use Cases (HLUC): - Monitoring and metering of infrastructure, energy consumption and water consumption - Efficient and sustainable energy usage - Fault detection and optimal maintenance scheduling - Reduction in grid congestion via grid reconfiguration Primary Use Cases (PUC): - Deployment of IoT sensors and monitoring equipment across the farm to monitor the status of the soil and weather conditions - Deployment of solar powered water pumps for day-time pumping and storage for night time pumping - Deployment of smart meters and energy management systems to schedule purchase of electricity from grid at cheaper hours, periods of low irradiation and facilitate export of excess electricity as a prosumer
Viewpoint	Technical
Type	HLUC

Table 24: Overview of the Agriculture Energy Hub at Igualada use case

Table 25: Short and Complete Description of the Igualada Smart Irrigation Use Case

Short Description	Igualada aims to modernise its agricultural irrigation infrastructure by integrating renewable-powered water pumps, smart meters, IoT soil sensors, and intelligent scheduling systems. Farmers will use real-time price signals, weather forecasts, and soil-condition data to optimize irrigation and energy consumption. This will reduce operating costs, enhance water efficiency, and support the transition to a sustainable agricultural ecosystem powered by renewable energy. Objective: Enable farmers to reduce energy and water consumption by introducing smart irrigation systems powered by renewable energy and storage while being managed through IoT-based monitoring, automated control, and price-responsive scheduling.
Complete Description	Historically, farmland in Igualada has been irrigated using fixed daily schedules derived from generic seasonal patterns. This approach fails to account for field-level variability such as differing soil types, uneven sun exposure, or localized drying rates. At the same time, irrigation has depended entirely on grid electricity, leaving farms exposed to fluctuating day-night energy prices, grid maintenance, and unexpected outages. The lack of real-time operational data has also prevented early detection of leaks or malfunctioning pumps and sprinklers, resulting in unnecessary water loss, increased costs, lost man hours in fault finding and potential regulatory penalties. To overcome these limitations, the farm deploys an AI-driven smart irrigation and energy management system. The solution integrates renewable-powered pumps, on-site battery storage, and a network of IoT sensors across the farmland. Soil moisture sensors and environmental monitors continuously capture field conditions, while smart meters track water and energy consumption in real time. This data is fed into an intelligent farm management platform that uses AI to analyze patterns and optimize decisions. The platform incorporates short- and medium-term weather forecasts alongside real-time and forecasted electricity price signals. Using these inputs, the AI dynamically determines optimal irrigation patterns for each zone of the farm. Irrigation cycles are automatically scheduled only when soil conditions indicate actual water demand, weather forecasts confirm favorable timing, renewable energy generation is available, and electricity prices are low. This ensures water is applied precisely where and when it is needed, avoiding over-irrigation and reducing energy costs. Energy storage plays a central role in the system’s optimization strategy. Batteries can be charged overnight when grid electricity prices are lowest, or during periods of high on-site solar generation. During the day, stored energy can be used to power irrigation pumps when prices rise or solar output fluctuates. When irrigation needs are already satisfied, based on soil data and upcoming weather forecasts, and both solar generation and storage capacity are in excess, the system can export electricity back to the grid during high-price periods, creating an additional revenue stream. Farmers interact with the system through dashboards, alerts, and AI-generated recommendations that explain irrigation and energy decisions in clear terms. They can override or fine-tune irrigation plans when necessary. The continuous sensor coverage also enables early identification of leaks, abnormal water flows, or faulty equipment, allowing issues to be resolved quickly. Overall, this use case demonstrates how AI-based irrigation patterns, combined with weather forecasts, electricity price signals, renewable generation, and energy storage, can significantly reduce water waste, lower operating costs, improve energy resilience, and enhance sustainability and crop yield consistency in Igualada.

Table 26: High-Level Use Cases, Related PUCs, and Relevant Actions for Igualada Smart Irrigation

High-Level Use Cases (HLUC)	• Real-time monitoring of soil moisture, temperature, and irrigation flow to maximise growing conditions for crops and increase crop yield • AI algorithms for automated irrigation based on soil measurements and weather forecasts to reduce manual labour • Renewable energy-powered pump operation (solar and battery based pumps) • Scheduling irrigation according to dynamic energy prices and forecasted renewable production • Smart metering for water and electricity usage • Predictive planning for water demand and energy costs
Relation to Other Use Cases (PUCs)	• Deployment of IoT environmental sensors across farmland (soil moisture, pH, weather stations) • Integration of renewable energy microgeneration (solar PV or wind turbines) to power pumps • Development of real-time energy-balancing algorithms for off-grid and hybrid systems • Connection of smart meters to irrigation controllers and farm management platforms • Utilization of predictive analytics to adjust irrigation patterns based on agriculture needs
Relevant Actions	• Installation of renewable-powered pumps and co-located storage systems • Deployment of IoT sensors and smart meters • Weather and price-forecast integration • Automated irrigation scheduling • Monitoring and continuous optimization

Business Layer

Identify and describe the main objectives, services, and stakeholders relevant to the scenario. What are the business goals, and who are the main actors?

Objectives

- Reduce irrigation operating costs
- Improve water usage efficiency
- Promote renewable energy adoption in agricultural operations
- Improve farm resilience through redundancy
- Increase farm productivity through automated decision-making
- Provide an additional revenue stream for the farm

Services

- Automated irrigation scheduling
- Energy and water usage analytics
- Price-responsive energy management
- Resource monitoring and reporting
- Compliance with agricultural water policies

Stakeholders

- Farmers and agricultural cooperatives (Prosumers and aggregators)
- Local energy providers (DSO)
- Municipal water authorities (Utilities)
- Renewable technology suppliers (Manufacturers)
- IoT and platform service providers (EMS, SCADA, SGAM)

Business Goals

- Lower energy bills through price-adaptive irrigation
- Reduce water waste through precise, sensor-driven control
- Support renewable energy integration within agricultural systems
- Improve overall sustainability of farmland irrigation

Actors

- Prosumer (farmer), Aggregator, DSO
- SCADA Supervisory Control and Data Acquisition
- Energy Management System (EMS)
- GIS Geographic Information System and Weather forecasting
- SGAM Smart Grid Architecture Model

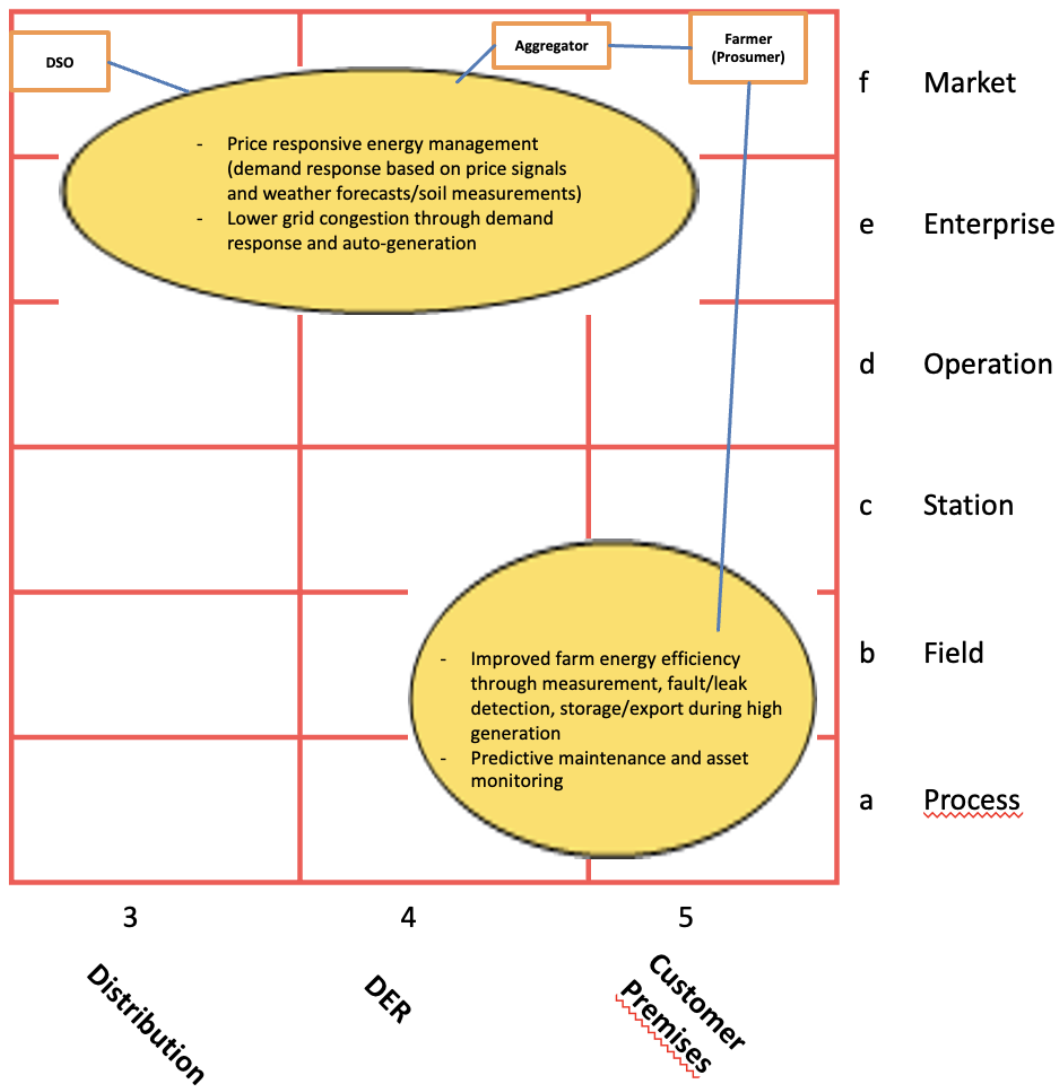


Figure 26: SGAM Business Layer Diagram

Function Layer

Detail the primary functions and operations required to achieve the business objectives. What processes and tasks need to be performed?

Primary Functions

1. **Environmental Monitoring:** Soil moisture, pH, temperature, humidity
2. **Demand Forecasting:** Predict water needs using weather and soil data
3. **Energy Forecasting:** Predict renewable production and energy prices allowing farmer to decide on grid import/export and/or storage in the BESS
4. **Irrigation Automation:** Control pumps and valves according to schedule
5. **Smart Metering:** Track electricity and water consumption
6. **Optimization:** Minimize energy costs and water use

Processes and Tasks

- Collect real-time soil and weather data via IoT sensors
- Analyse weather forecasts (rain probability, temperature, evapotranspiration)
- Predict irrigation intervals and locations around the farm
- Activate pumps based on thresholds and price signals (choosing to use PV installed on pumps, energy stored in BESS and/or grid electricity)
- Monitor pump health and energy generation
- Provide reports and optimization insights to farmers

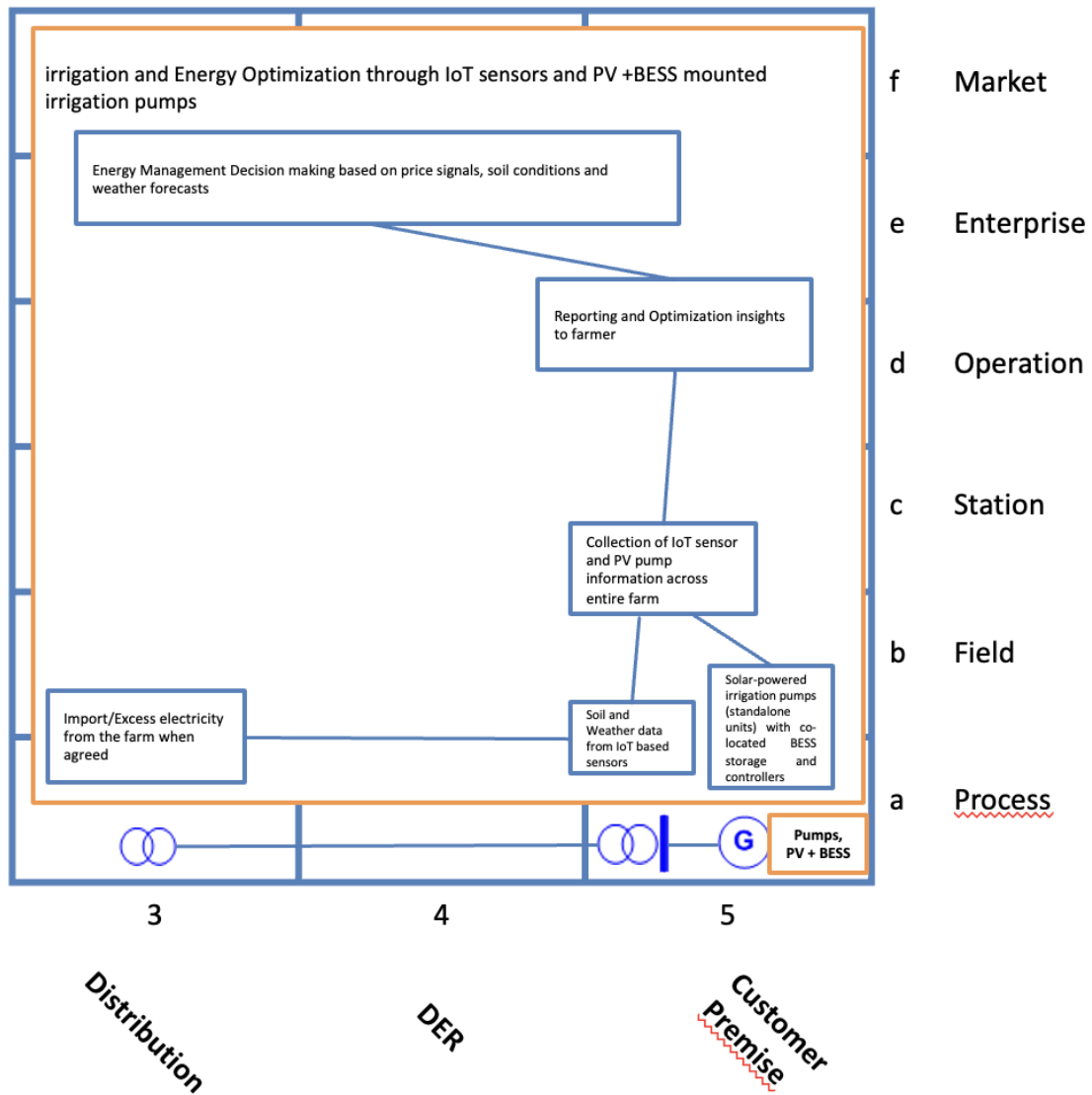


Figure 27: SGAM Functional Layer Diagram

Component Layer

Identify the physical components and devices relevant to the scenario. What are the main pieces of infrastructure or equipment?

Physical Components

- Solar-powered irrigation pumps (standalone units) with co-located BESS storage
- MV switchgear and medium-voltage cable network
- IoT sensors (soil moisture, pH, temperature, weather stations)
- FO cabling and central RTU
- Smart meters (water and electricity)
- Battery storage systems
- Irrigation controllers and actuated valves
- Farm EMS (Energy Management System)
- Communication routers and gateways

Key Infrastructure

- Intelligent pump systems fitted with PV panels and storage
- MV cable network and switch house for connection to distribution network
- Sensor networks across fields
- Smart metering infrastructure
- Farm-level SCADA or cloud-based irrigation management platforms

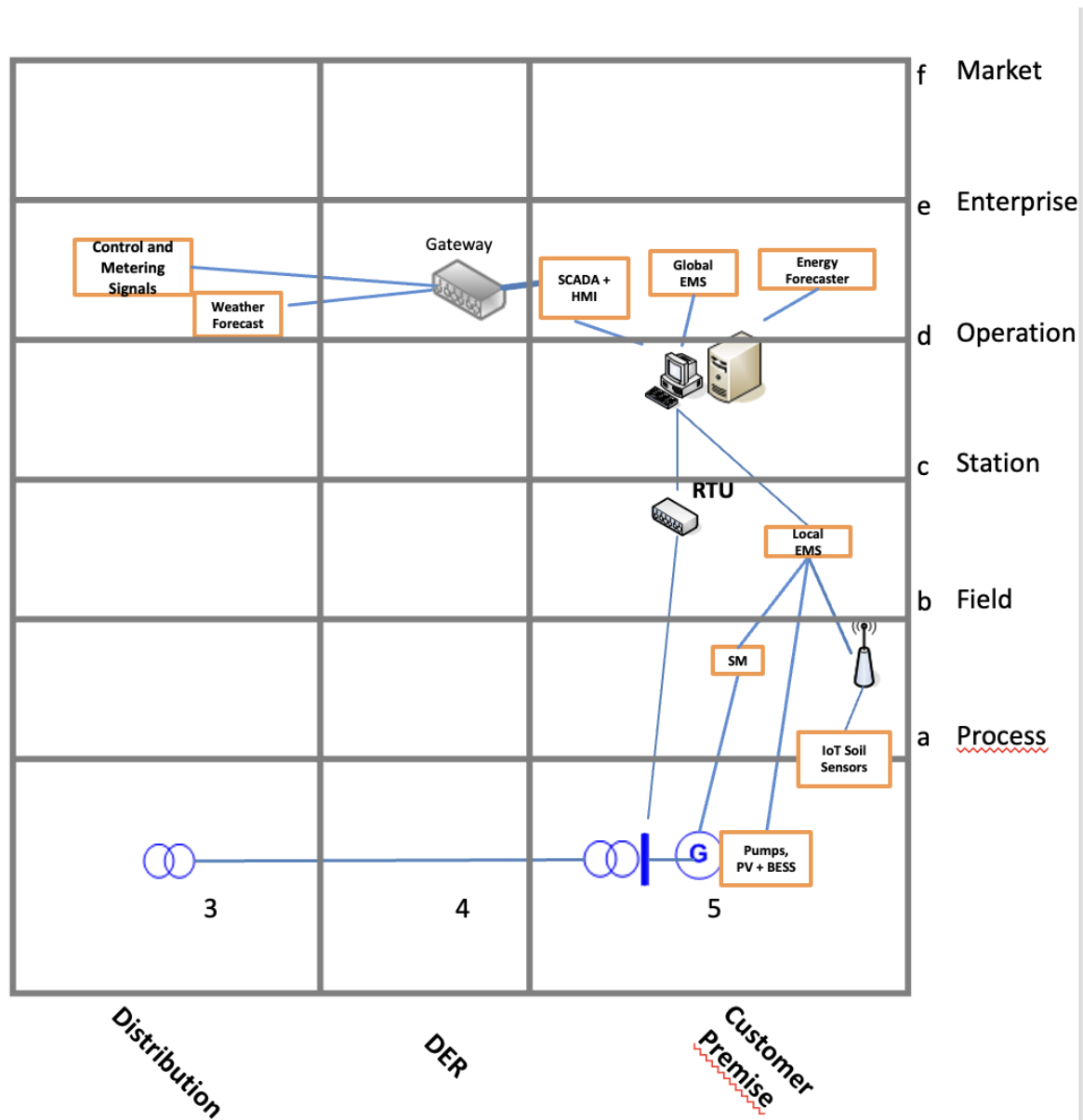


Figure 28: SGAM Component Layer Diagram

Information Layer

Detail the types of data and information that will be processed and shared. What data formats, standards, or schemas are relevant?

Types of Data

- **Real-Time Data:**

- Electrical parameters of irrigation pumps (voltage, current, power)
- Soil measurements of moisture, pH, temperature
- Water flow rates, pressure, soil moisture levels, and pump temperatures
- Renewable generation and BESS data (irradiation, module temperature, SOC of BESS)

- **Predictive Alerts:**

- AI-based alerts for pump failure, pipe leaks, abnormal water consumption, or energy inefficiency
- Irrigation scheduling alerts based on weather forecasts and energy price predictions

- **Maintenance Records:**

- Historical maintenance logs for pumps, valves, pipelines, meters, and renewable energy assets
- Records of repairs, inspections, and equipment upgrades

- **Water and Energy Flow Data:**

- Water withdrawal, distribution, and consumption data per field or farm
- Energy consumption of irrigation systems and on-site renewable energy production
- Interaction between irrigation demand and electricity grid or local microgrids

- **Farmer and Consumer Data:**

- Irrigation usage profiles per farm
- Demand-response participation logs (load shifting irrigation to low-price periods)
- Pump PV generation, BESS State of Charge, price signals for exporting/importing electricity

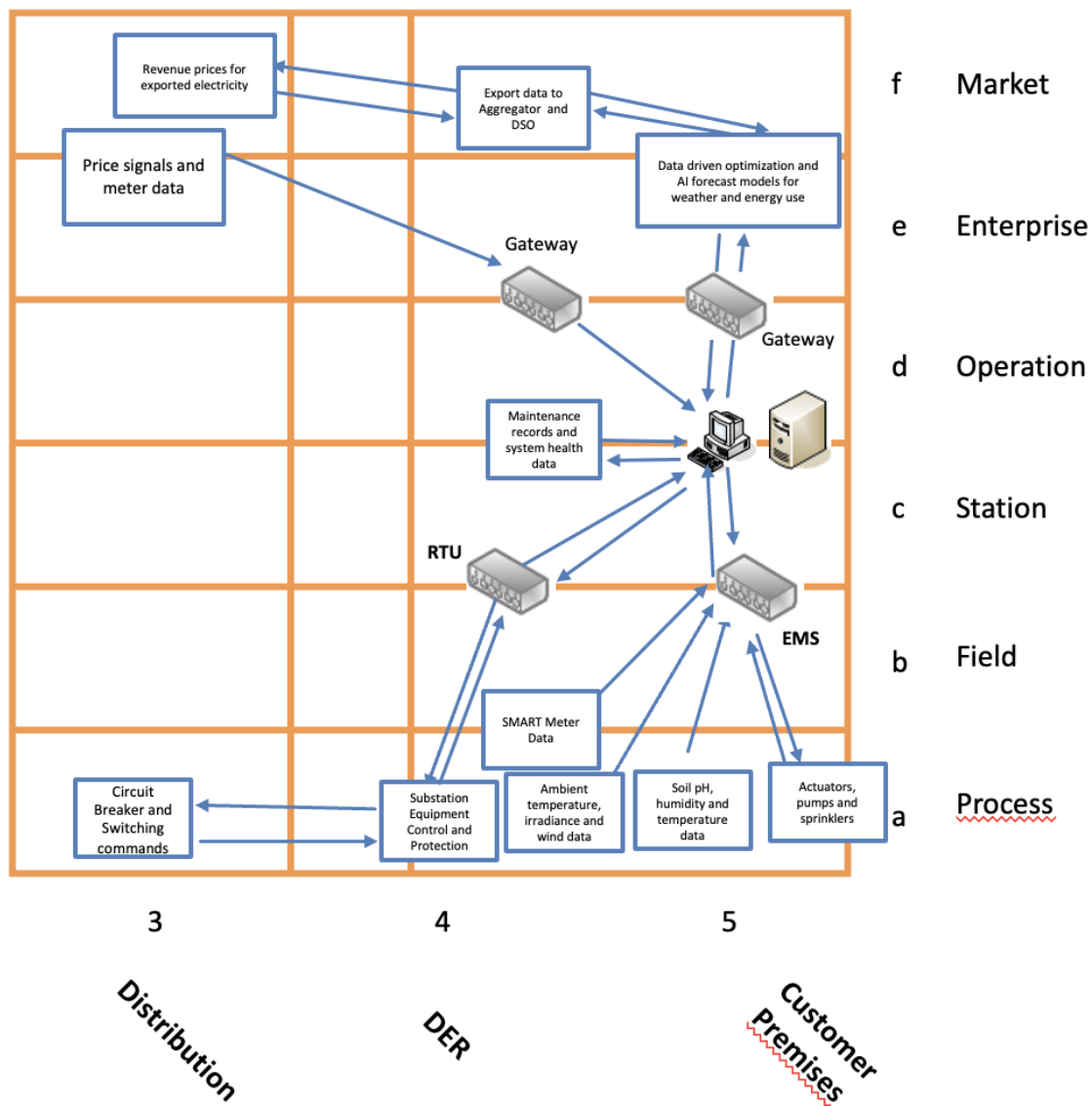


Figure 29: SGAM Information Layer Diagram

Communication Layer

5.13 Data Formats and Standards

- **IEC 61850**
Applied where smart substations, renewable plants, or grid-connected pump control systems are present.
- **DLSM/COSEM**
Used for smart meter data transfer to the local EMS.
- **CIM (Common Information Model)**
Used for energy data exchange between utilities, aggregators, and market platforms.
- **IoT and Agricultural Data Formats (Zigbee/MQTT)**
Used for communication between soil sensors, smart meters, pump controllers, and central platforms.

5.14 Key Considerations

- **Data Accuracy and Timeliness**
Reliable real-time data is essential for optimizing irrigation schedules based on energy prices and weather conditions.
- **Interoperability**
Standardized data models enable seamless integration of energy systems, irrigation infrastructure, weather services, and market platforms.
- **Security and Privacy**
Farm operational data, energy usage profiles, and market participation data are protected using encryption, authentication, and access control mechanisms.

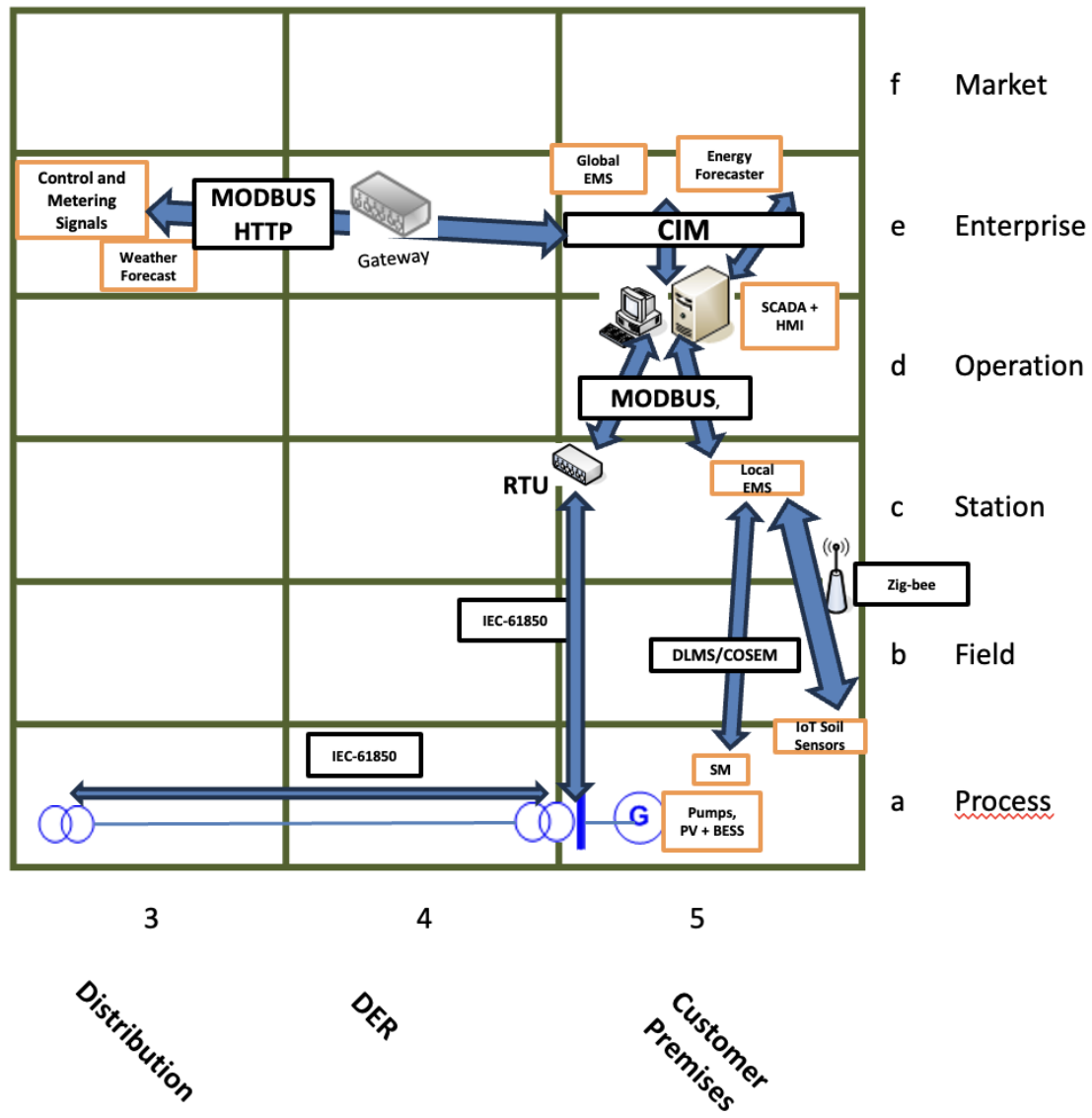


Figure 30: SGAM Communication Layer Diagram

6 Conclusion

This report discussed the analysis and application of smart grid methodology to a hypothetical dysfunctional transmission grid in Catalonia. Red Eléctrica data was used to create realistic 24 hour generation and demand profiles, which were applied to a base case grid simulation in PandaPower. Based on the identified shortcomings of the base case grid, various solutions were proposed involving extra transmission lines and modern power generation and storage technologies. An economic estimate was made for each case to conclude on the most feasible solution for the ailing grid. While each solution was able to provide savings on operating costs, the ideal solution economically would be to implement renewable energy generation and demand response programming (Solutions 1 and 3). The extra generation is cheap enough to be economically beneficial, and the demand response also reduces energy import costs without the excessive equipment expenses of Solution 2. Additionally, the Smart Grid Architecture model was applied to a use case for smart irrigation and farm management around Igualada. The necessary information system topology was analyzed in detail to prepare for the hypothetical implementation of smart grid technology.

References

- [1] Agency for the Cooperation of Energy Regulators (ACER). *Report on Unit Investment Cost Indicators and Corresponding Reference Values for Electricity Infrastructure*. Technical Report. Ljubljana, Slovenia: Agency for the Cooperation of Energy Regulators, 2015. URL: https://www.acer.europa.eu/sites/default/files/documents/Official_documents/Publications/UIC_Electricity_History/UIC%20Report%20%20-%20Electricity%20infrastructure.pdf.
- [2] Yunus Biçen, Faruk Aras, and Hulya Kirkici. “Lifetime estimation and monitoring of power transformer considering annual load factors”. In: *IEEE Transactions on Dielectrics and Electrical Insulation* 21.3 (2014), pp. 1360–1367. DOI: 10.1109/TDEI.2014.6832284.
- [3] Mark Bolinger and Greta Bolinger. “Land Requirements for Utility-Scale PV: An Empirical Update on Power and Energy Density”. In: *IEEE Journal of Photovoltaics* PP (Jan. 2022), pp. 1–6. DOI: 10.1109/JPHOTOV.2021.3136805.
- [4] International Renewable Energy Agency. *Renewable Power Generation Costs in 2024*. Abu Dhabi: International Renewable Energy Agency, 2025.
- [5] Red Eléctrica de España (REE). *Generación y consumo (ESIOS)*. <https://www.esios.ree.es/es/generacion-y-consumo?date=02-01-2026>. Accessed 11 Nov 2025. 2026.
- [6] Vestas Wind Systems A/S. *V162-6.2 MWTM*. 2025. URL: <https://www.vestas.com/en/energy-solutions/onshore-wind-turbines/enventus-platform/v162-6-2-mw> (visited on 01/03/2026).